REAL-TIME FEDERATED LEARNING FRAMEWORK FOR DETECTING CARDIOVASCULAR DISEASES IN ECG SIGNALS

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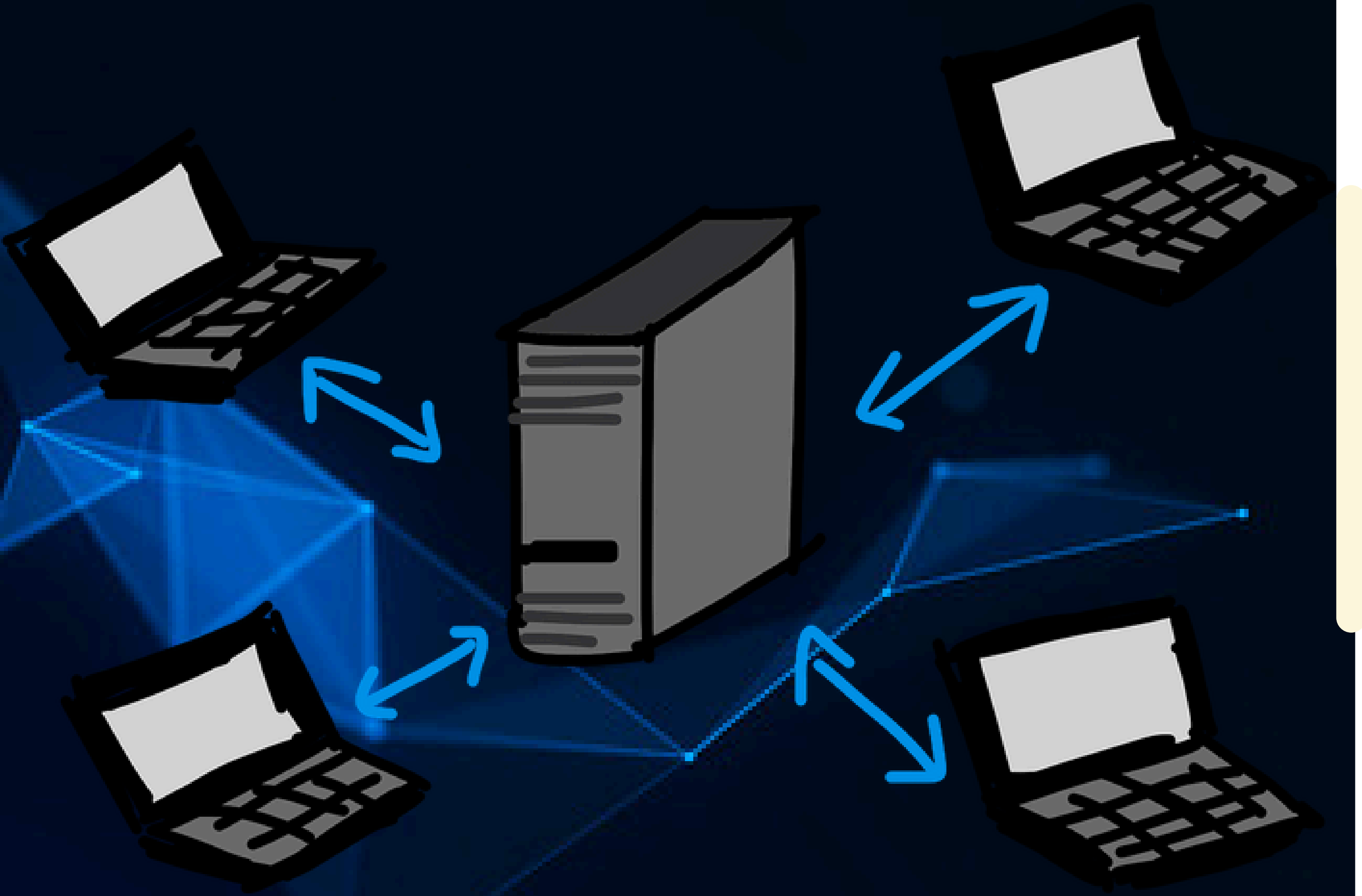
INTRODUCTION

The use of ECG, or also known as Electrocardiograms, can represent the electrical behaviour of the heart and in parallel, are employed as diagnostic tools for detecting arrhythmias, unsystematic variations of the cardiac pulsations.

However, despite this methodology of analysis is widely use in clinical instances, according to [1], heart diseases are becoming one of the frequent causes of death in the United States. Every 33 seconds a person lose their live due to this condition and according to a clinical analysis performed in 2022, every 5 deaths, one of them is due to a heart affectation.

WHY FEDERATED LEARNING?



Following the study presented and considering the frequency that heart diseases are becoming even more recurrent in the nowadays, these aspects motivated the development of the project in question, which addresses the detection of six different types of arrhythmias, given an input ECG signal using a real-time Federated Learning (FL) strategy.

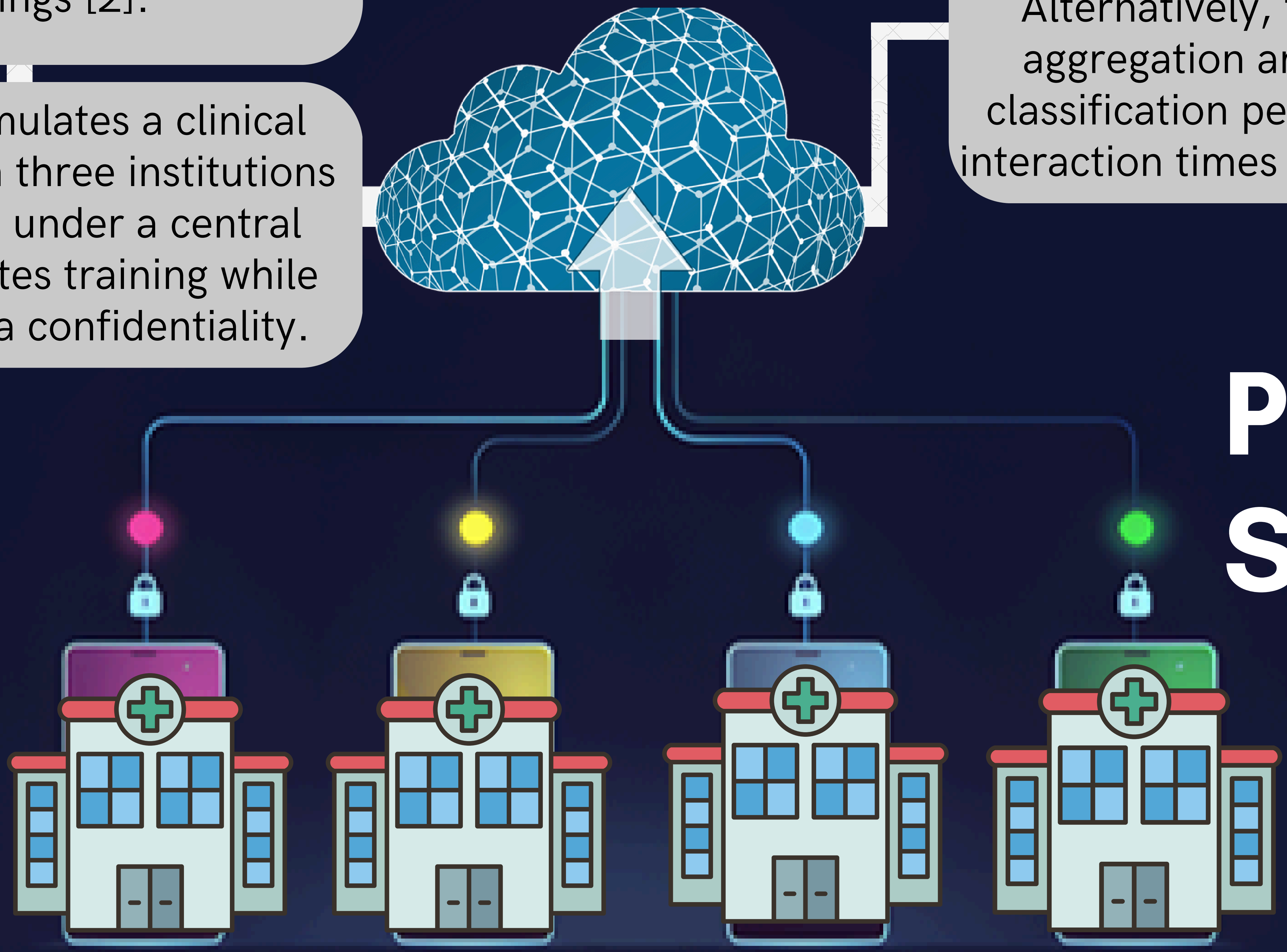
This framework allows that multiple clients can train different models of classification, using their local data without exposing sensitive information or affecting the confidentiality of the patients' clinical history. Additionally, in the same time, the models contribute to build a set of common weights that better satisfy the robustness on detecting effectively the six classes given, generating minimal delays in updates, low-latency interactions with server and rapid adaptation to new data.

PROPOSED SCHEME

The proposed framework consists on the designing of a real-time FL system capable of providing the classification of six cardiovascular diseases from 12-lead ECG recordings [2].

The architecture simulates a clinical environment in which three institutions (clients) collaborate under a central server that coordinates training while preserving local data confidentiality.

Each client processes its own data and communicates only model parameters, enabling privacy-preserving collaboration across distributed datasets. Alternatively, three methods of model aggregation are compared in terms of classification performance, execution and interaction times between server and client.



METHODS OF AGGREGATION SELECTED

FedAvg

Consists of calculating the standard average of weights and biases of the clients assigning equal level of importance to each client in the aggregation.

$$G_w = \frac{1}{n\_clients} * \sum_{i=1}^{n\_clients} W_i$$

Performance-based [3]

Assigns more importance in the calculation of weights to those clients that performed better and achieved a higher score in the classification of the local test sets.

$$\phi_j = \frac{1}{\sum_{k=1}^{n\_clients} \frac{1}{acc_{jk}}} * \sum_{i=1}^{n\_clients} \frac{1}{acc_{ji}} * W_{ji}$$

DQ-Fed [4]

Based on the calculation of the balance of the classes seen on each local dataset, such as the presence of noise and the variation of loss, the models with better equity in labels and higher stability will have more influence in the aggregation.

$$\omega_{global} = \frac{\sum_{i=1}^{k} NP_k * SE_k^{norm} * \omega_k}{\sum_{i=1}^{k} NP_k}$$

METHODOLOGY- DATA PREPROCESSING

Dataset Analysis → Downsampling → Filtering

Dataset Analysis

- According to [2], the dataset contains ECG signals extracted from six different institutes and challenges across the world.
- Includes 27 different cardiovascular arrhythmias.
- The recordings vary from 6 seconds to 30 minutes, with sampling frequencies ranging 257 Hz up to 1 kHz.

Class	Number of Recordings
Atrial fibrillation (AF)	4,026
Bradycardia (Brady)	289
Prolonged QT interval (LQT)	2,253
Q wave abnormal (QAb)	1,252
Sinus Arrhythmia (SA)	1,476
Sinus Tachycardia (STach)	3,050

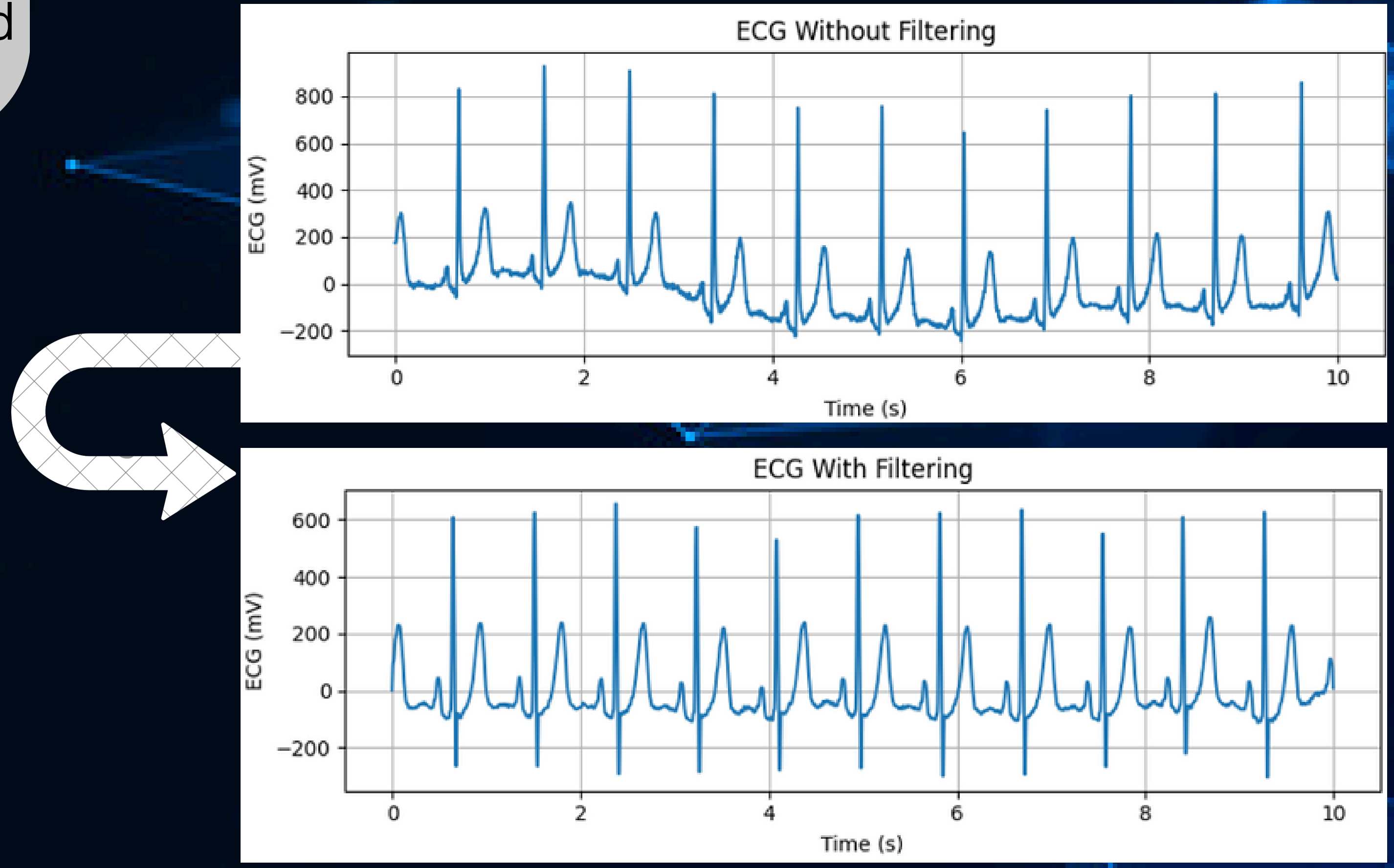
Downsampling

With the application of a downsampling, this ensures that all signals are consistent in terms of resolution or number of samples in a given interval of time, as well as it reduces the computational load in long-duration recordings.

For this, the sampling frequency selected was 250 Hz.

Filtering

A filter stage was required in order remove high frequency variations, noise and baseline drift, in which, this is achieved implementing a 7-order band-pass filter of 0.5 Hz as low cut and 45 Hz in high cut.



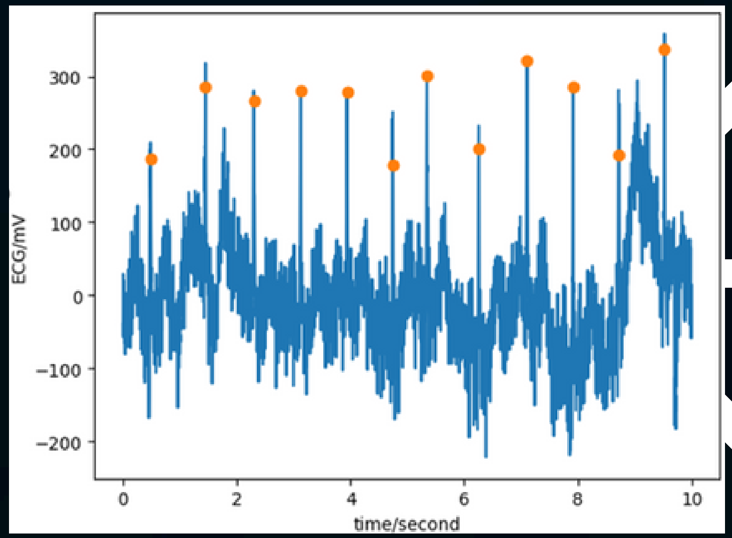
METHODOLOGY- DATA PREPROCESSING

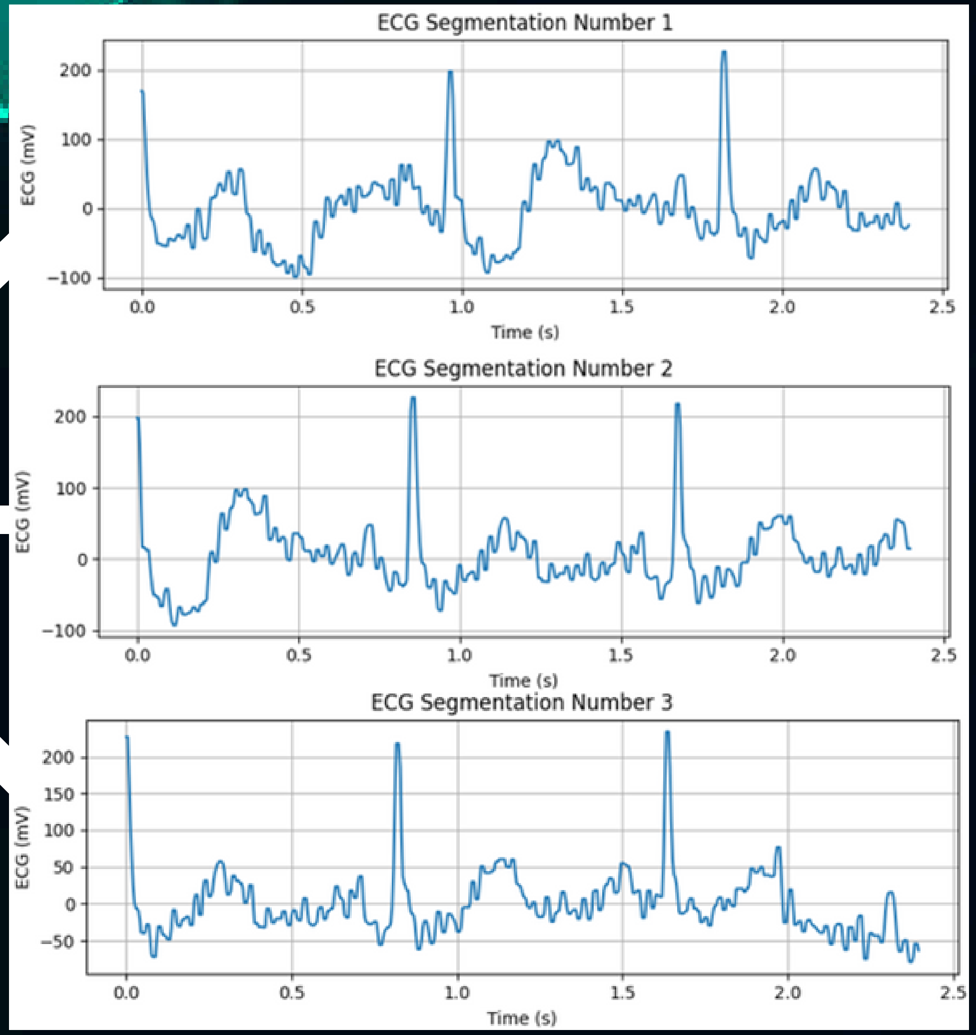
Segmentation

Feature Selection

Due to the high beat density in many ECG recordings, the article [5] reported that retaining at least three consecutive beats preserves arrhythmia morphologies and optimizes information retention, while enhancing model generalization and noise suppression. Accordingly, [6] a 2.3 second sliding window enables capturing three full cycles under low-frequency conditions.

Each ECG recording provided contains 12 lead, which correspond to different perspectives and combinations of the heartbeat measurements in a single sample. However, a portion of these can be redundant and dependent on other leads. Thus, as described in [7], 4 leads can be removed mainly because these are linear combination of the remaining leads, hence the information is implict already and does not provide any additional dimensionality or relevance in the analysis.





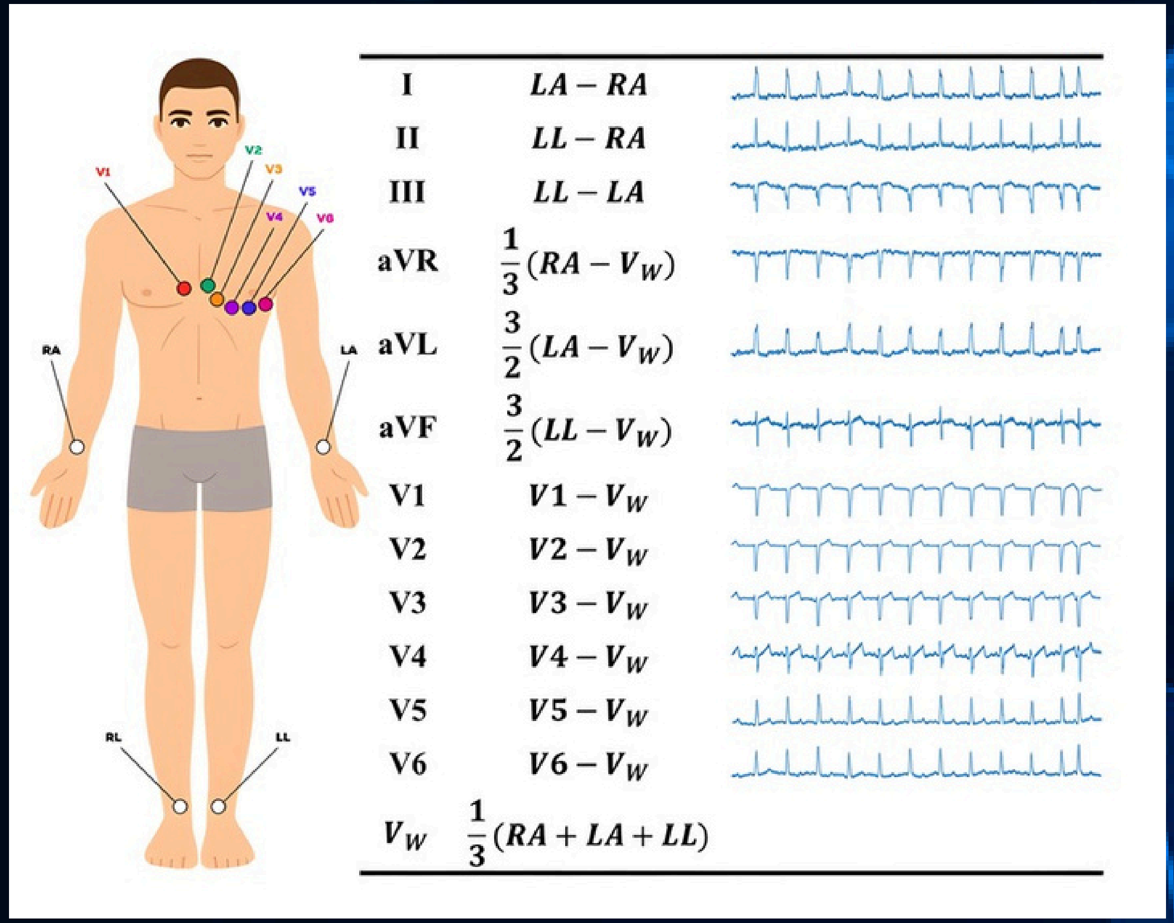


Image taken from [8]

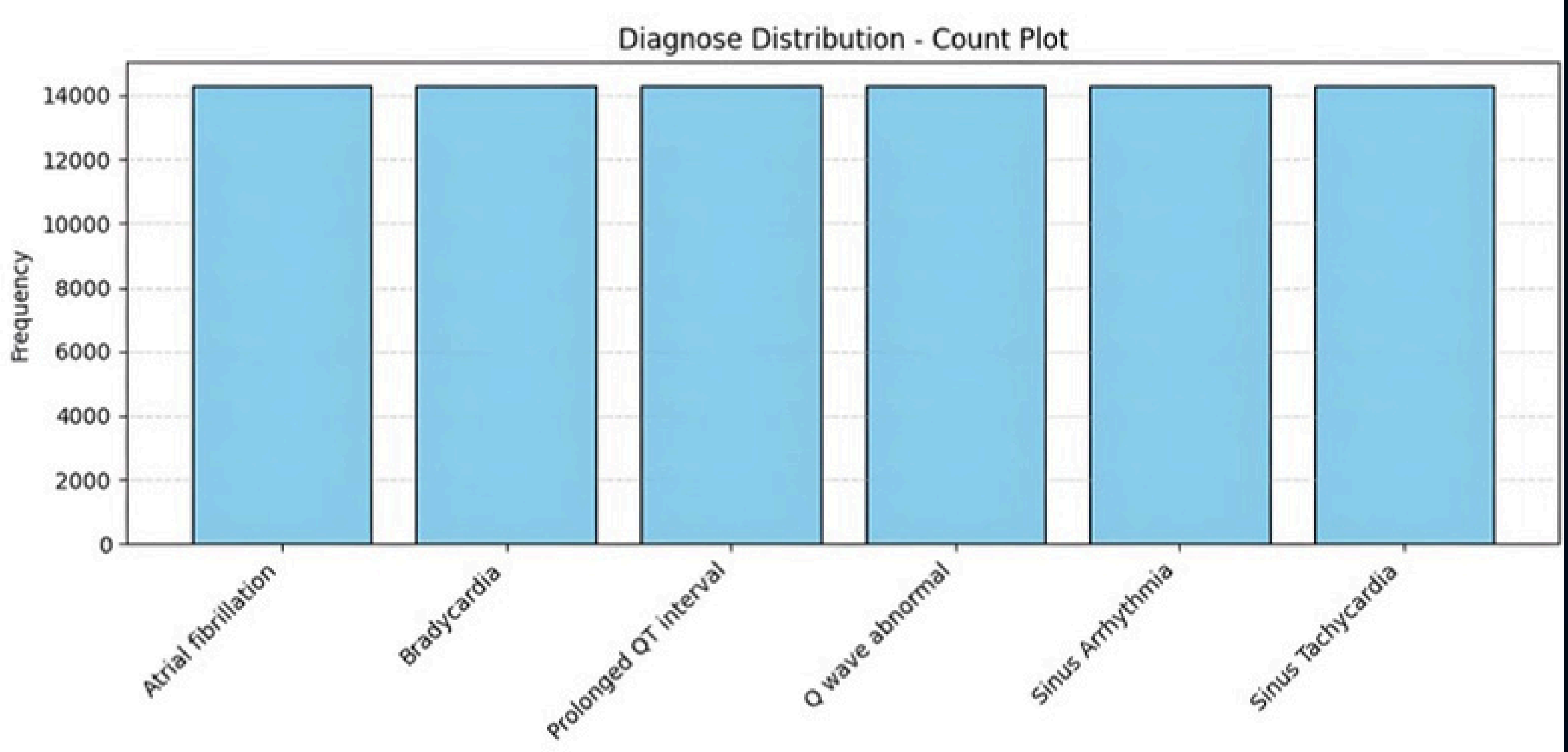
METHODOLOGY- DATA PREPROCESSING

Data Augmentation

Signal Scaling

Observing that class distribution is excessively imbalanced, and that the minority class only takes 4.0% of the total data, a proposal in order to address this condition was to limit the number of samples of the rest of classes to the second minority class, given that this one in particular has more than twice the number of samples of the minority label.

Once all the rest of classes have the same number of samples, a set of data augmentation techniques were applied to the minority class, these include the addition of low-scale random noise, amplitude reduction and waveform shifting.



The magnitudes seen in the ECG signals are given in mV, in which vary from either very large positive or negative numbers. This condition might affect the convergence of parameters and training times of the classifiers. For that reason, a minimum and maximum scaler was selected, given that, it maintains the morphology of the signal without modifying the patterns of the arrhythmias.

$$Scaled\ Array[i] = \frac{Data\ [i] - Min\ Value}{Max\ Value - Min\ Value}$$

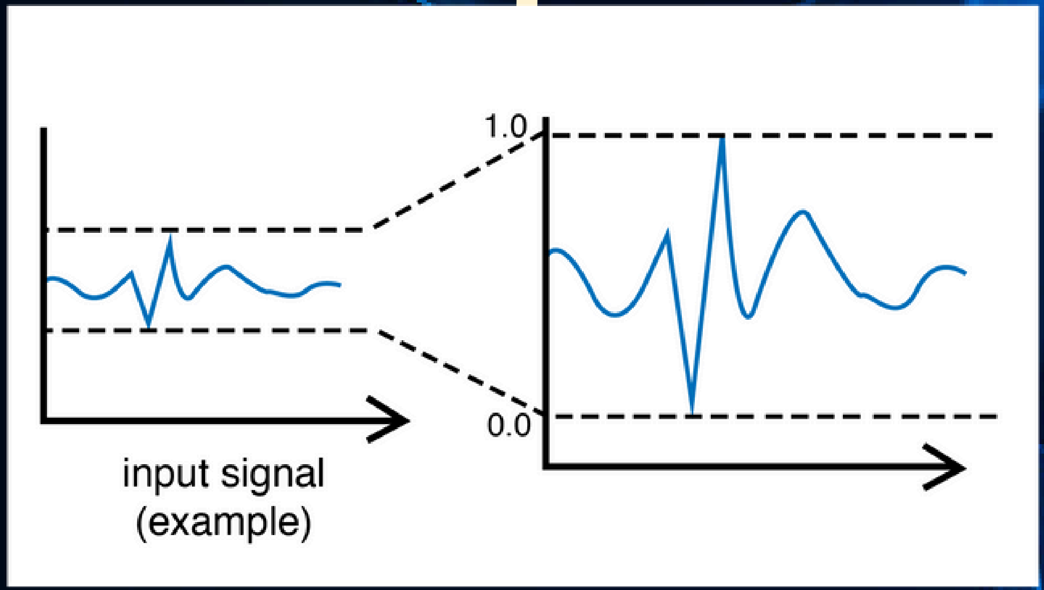


Image taken from [9]

DATASET SEPARATION

FIRST SPLIT

Since three clients will be deployed, the dataset is split to simulate a FL setup, where each institution keeps its own private data. Each client receives 30% of the preprocessed samples, while the remaining 10% is reserved for testing after training.

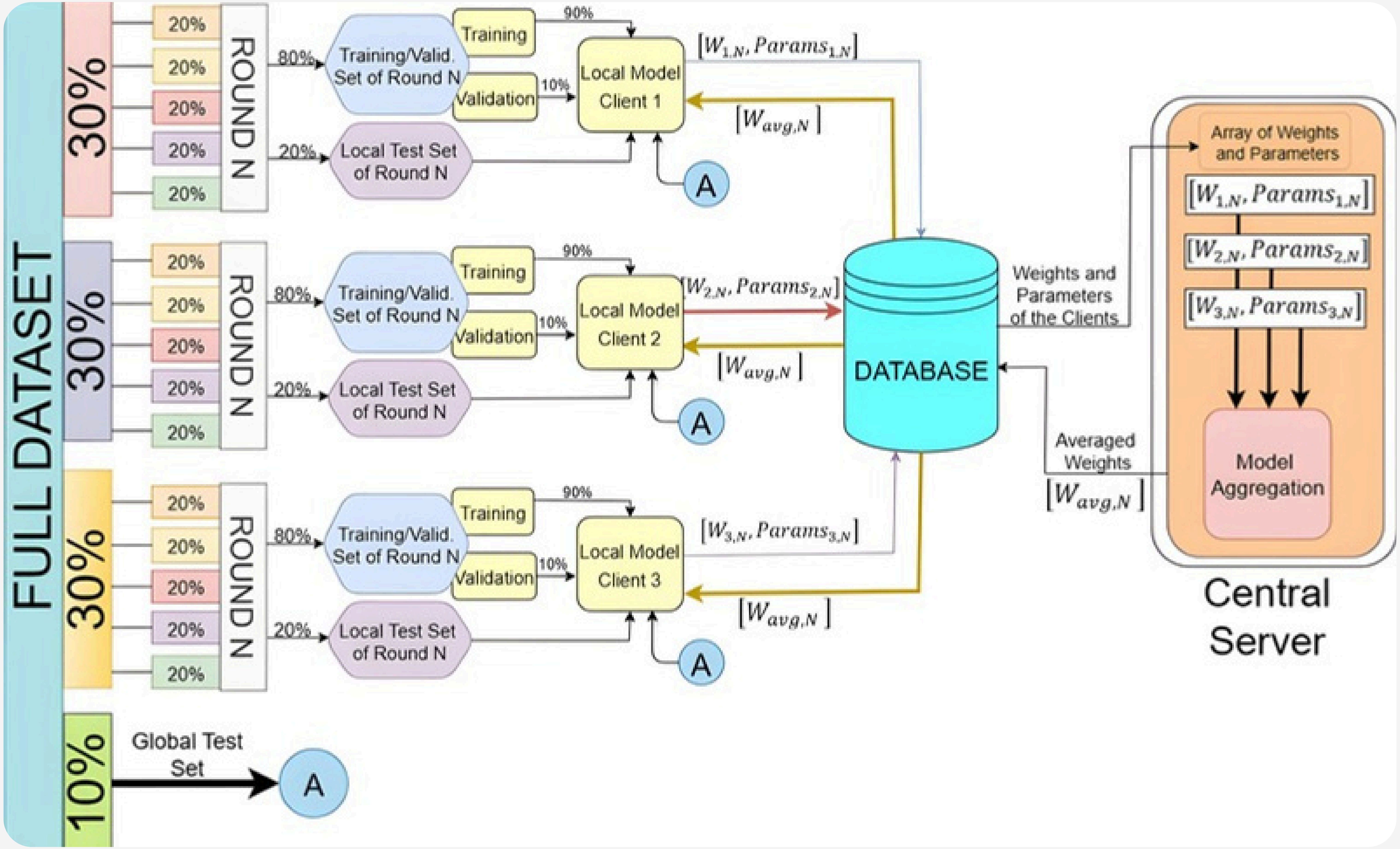
The data proportions were chosen to maximize the amount of data each local instance uses for training and hyperparameter tuning. The testing set includes a wide variety of samples from each class to properly evaluate how well the models can identify and distinguish different arrhythmias.

SECOND SPLIT

Each client is trained using incremental learning, where training is divided into stages instead of using all the data at once. This is done in five rounds, with each round using 20% of the dataset.

In each round, the data is further split: 80% for training/validation (with 90% training and 10% validation) and 20% for testing after the round ends. Once training and testing are completed, the client's model sends its weights and biases to the global server.

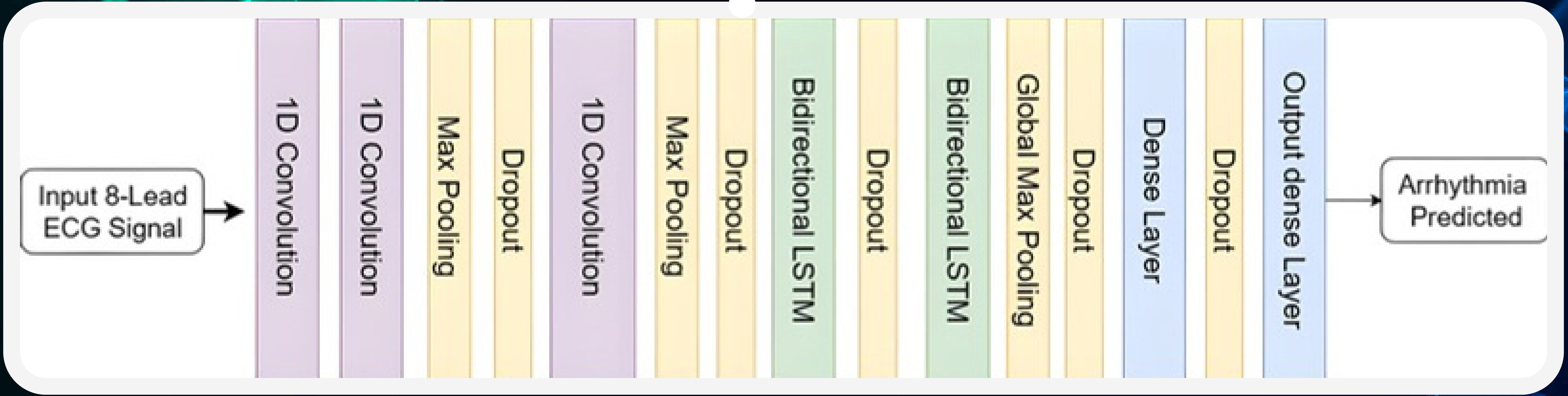
CLIENT - SERVER INTERACTION



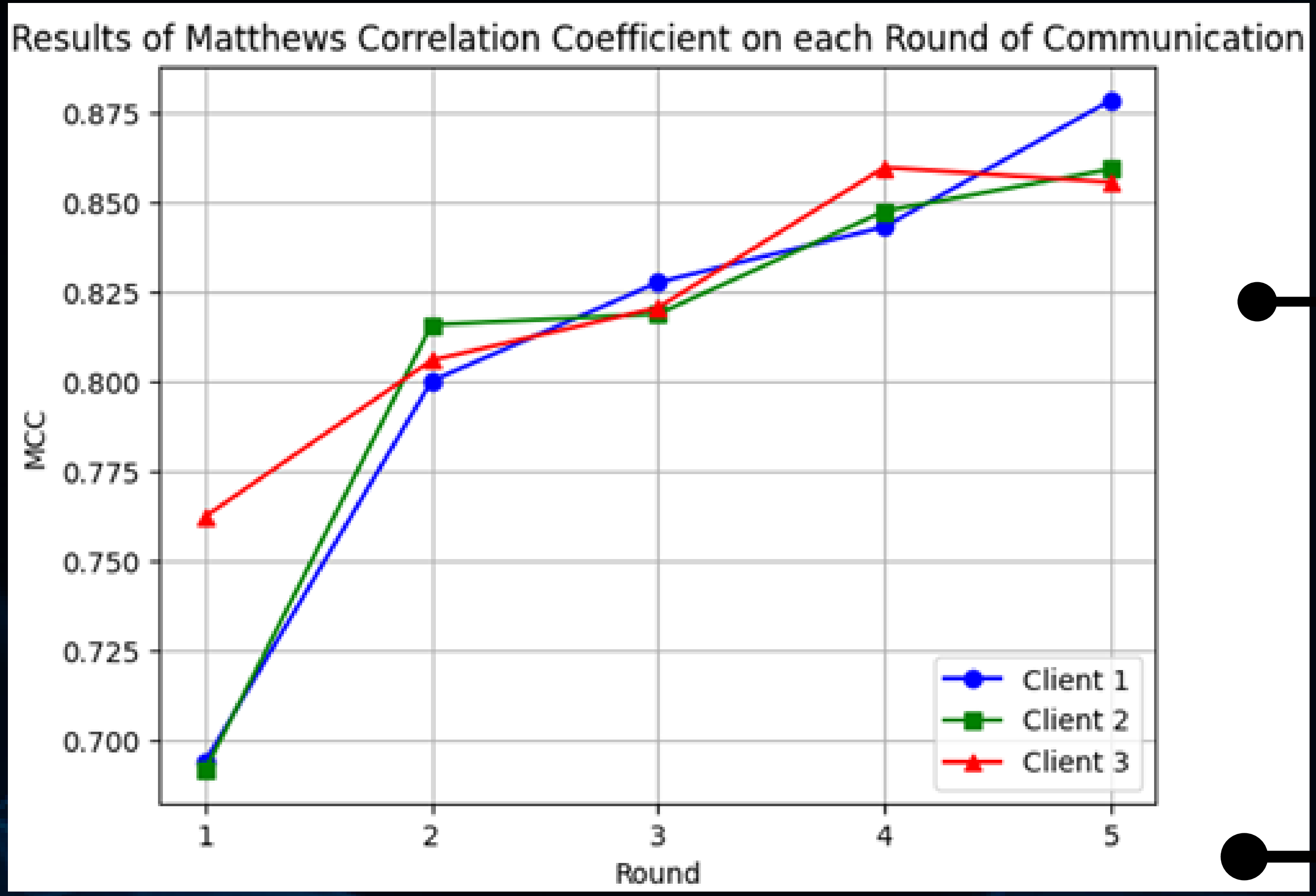
After each round, clients send their weights and parameters to the server via the database. The server aggregates them, updates the global weights, and saves them back. Then, each client retrieves the new weights, updates its model, and continues training.

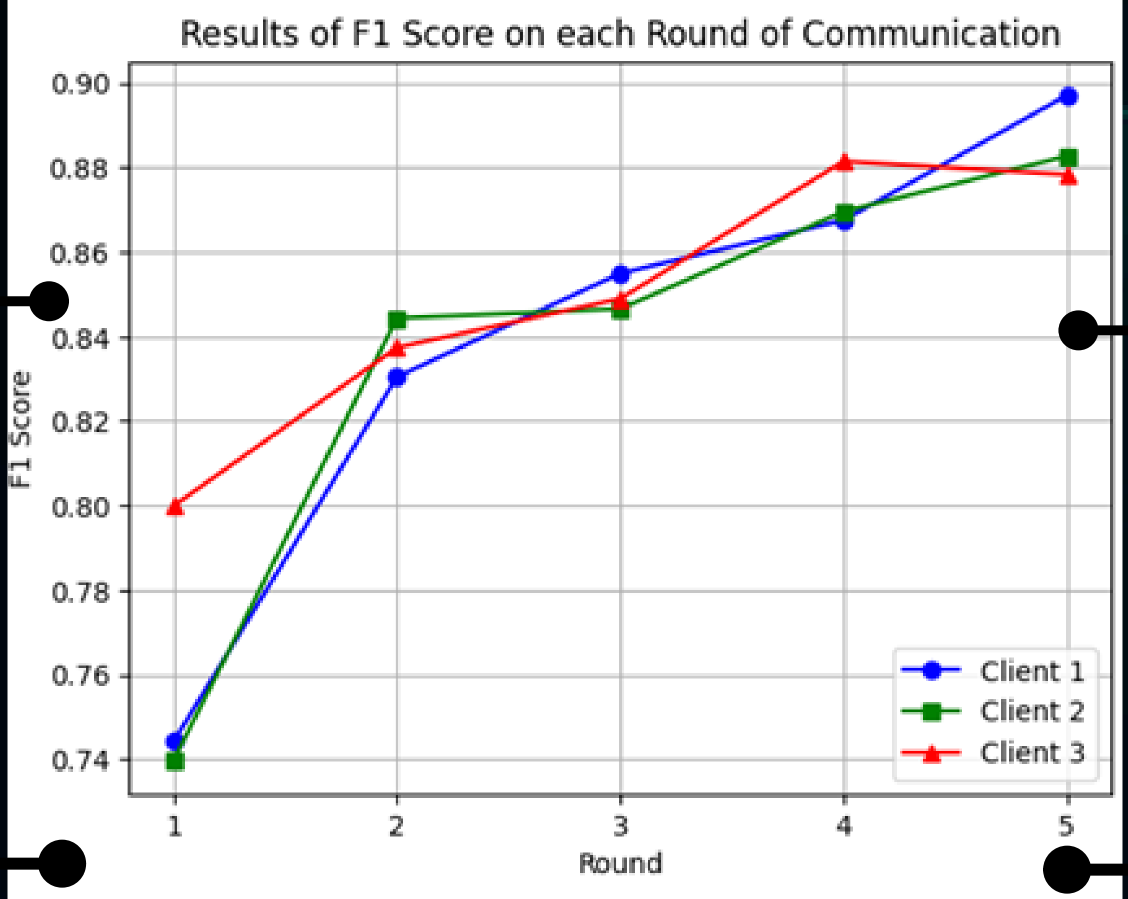
MODEL PROPOSED - 1D CNN BILSTM

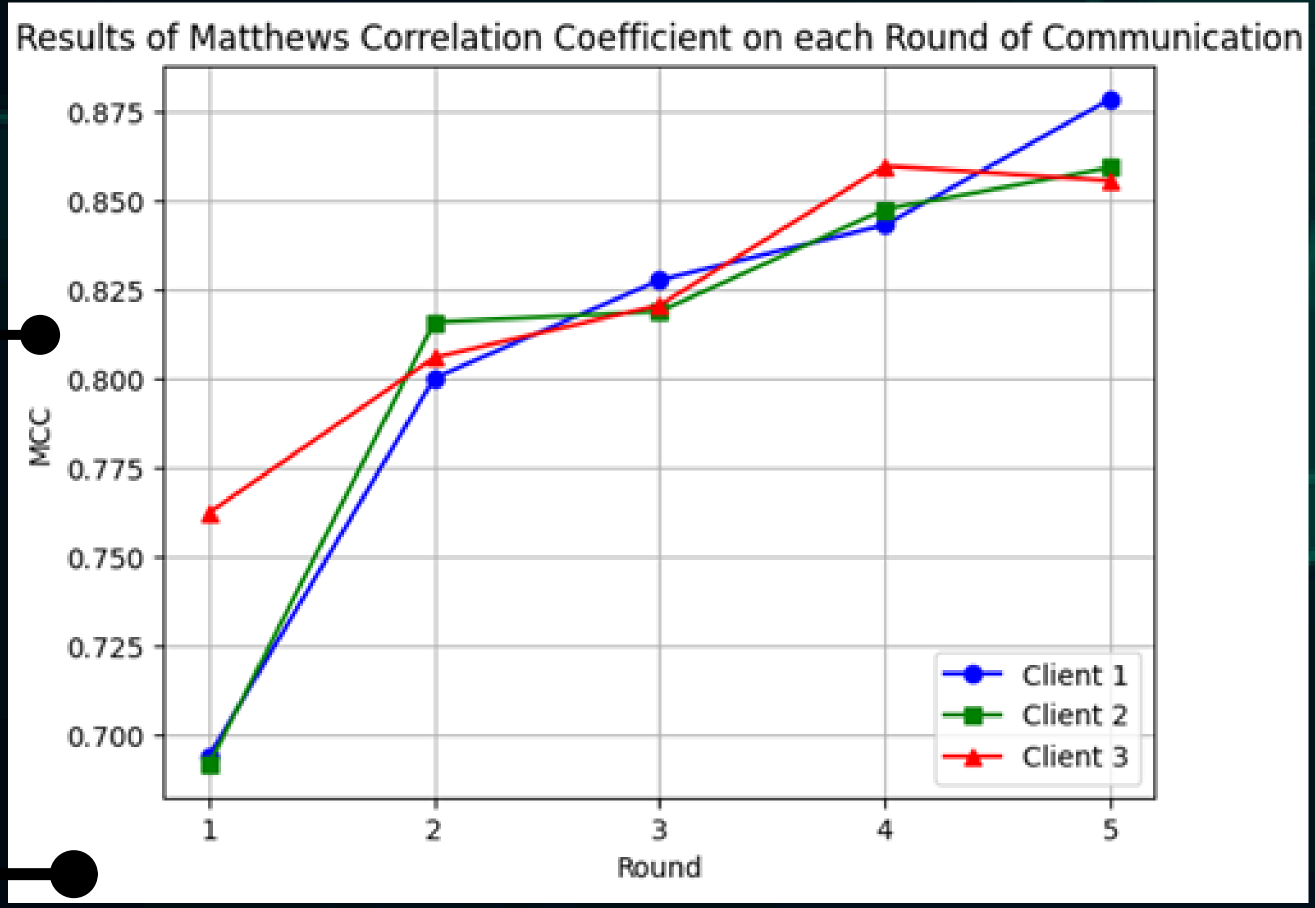
The model uses three 1D convolutional layers for feature extraction, followed by two BiLSTM layers to analyze temporal patterns in both directions. A dense layer with 128 neurons processes the features, and the output layer (6 neurons) provides confidence scores for each arrhythmia class.



RESULTS OF THE FRAMEWORK USING PERFORMANCE BASED AGGREGATION

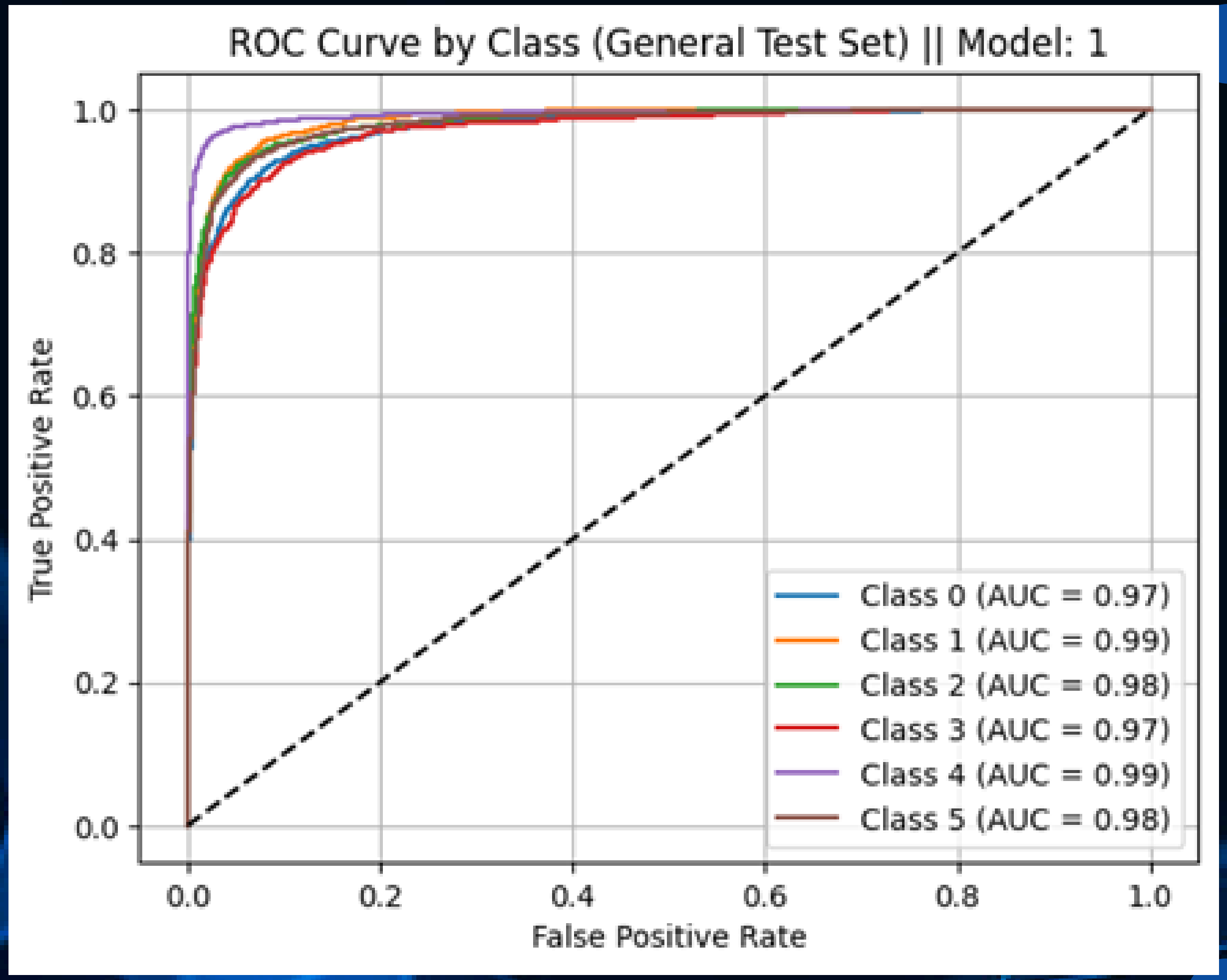


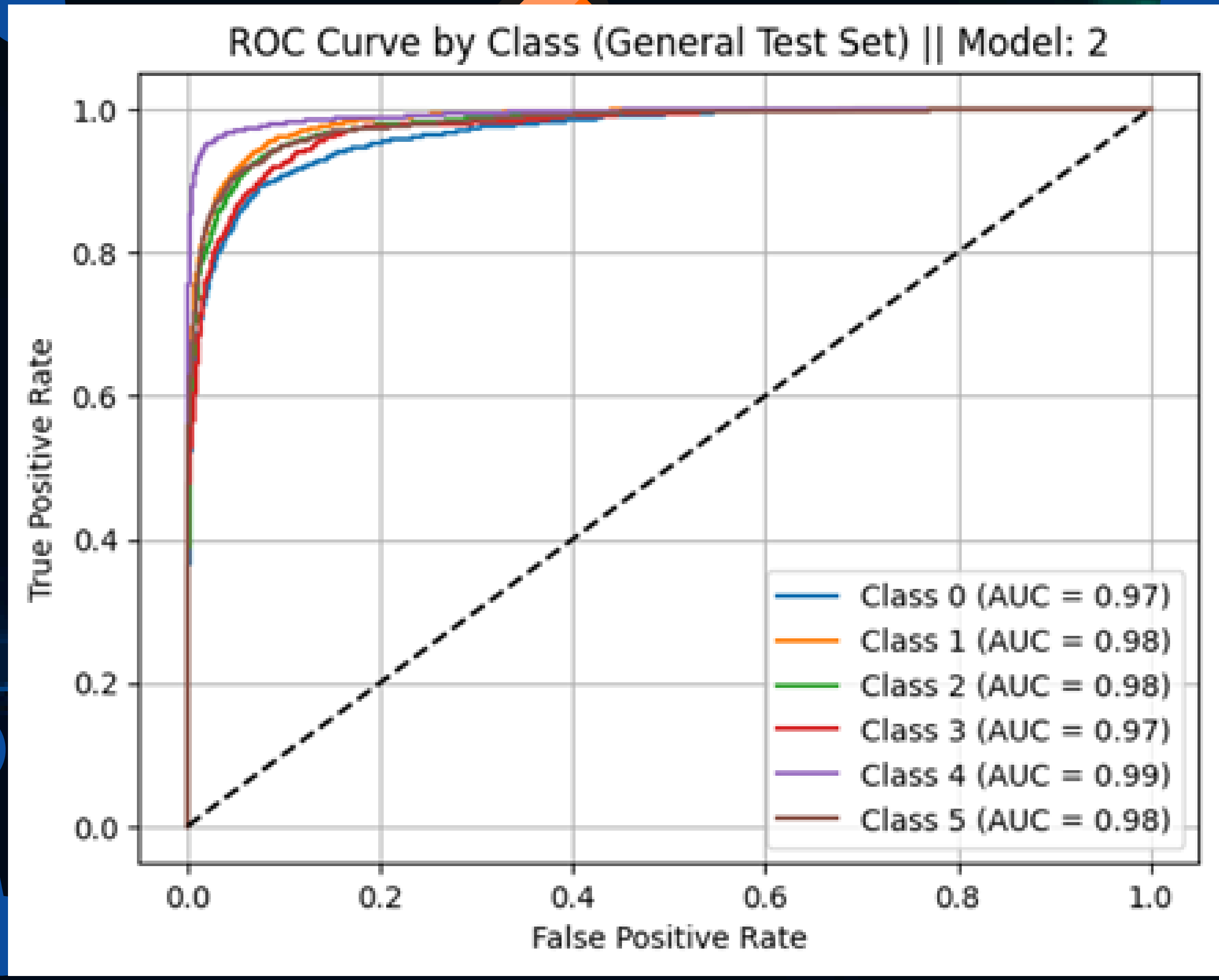


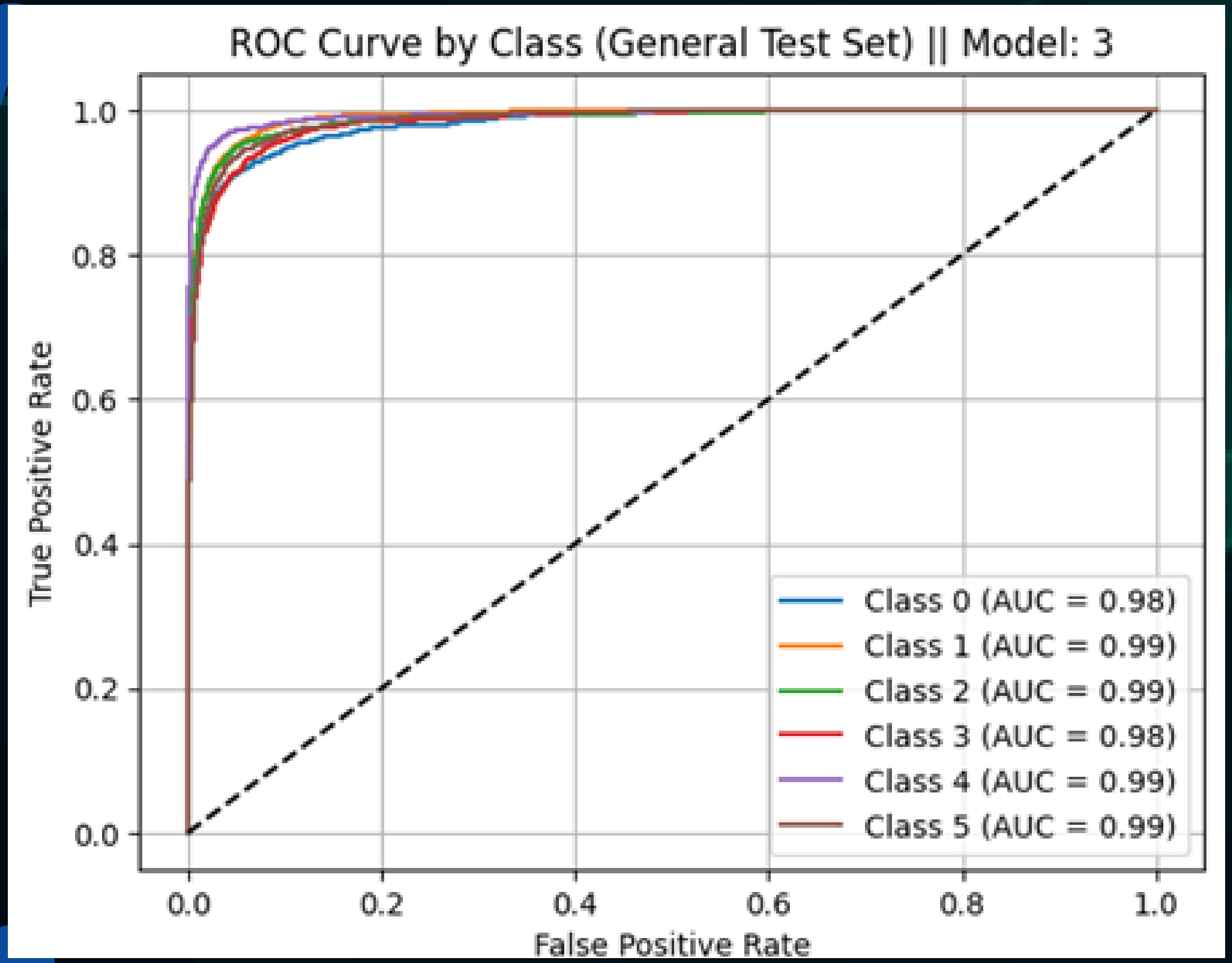


Client	Accuracy	F1-Score	MCC	Specificity
1	0,862	0,861	0,835	0,973
2	0,854	0,852	0,825	0,971
3	0,886	0,885	0,864	0,977
Average	0,867	0,866	0,841	0,974

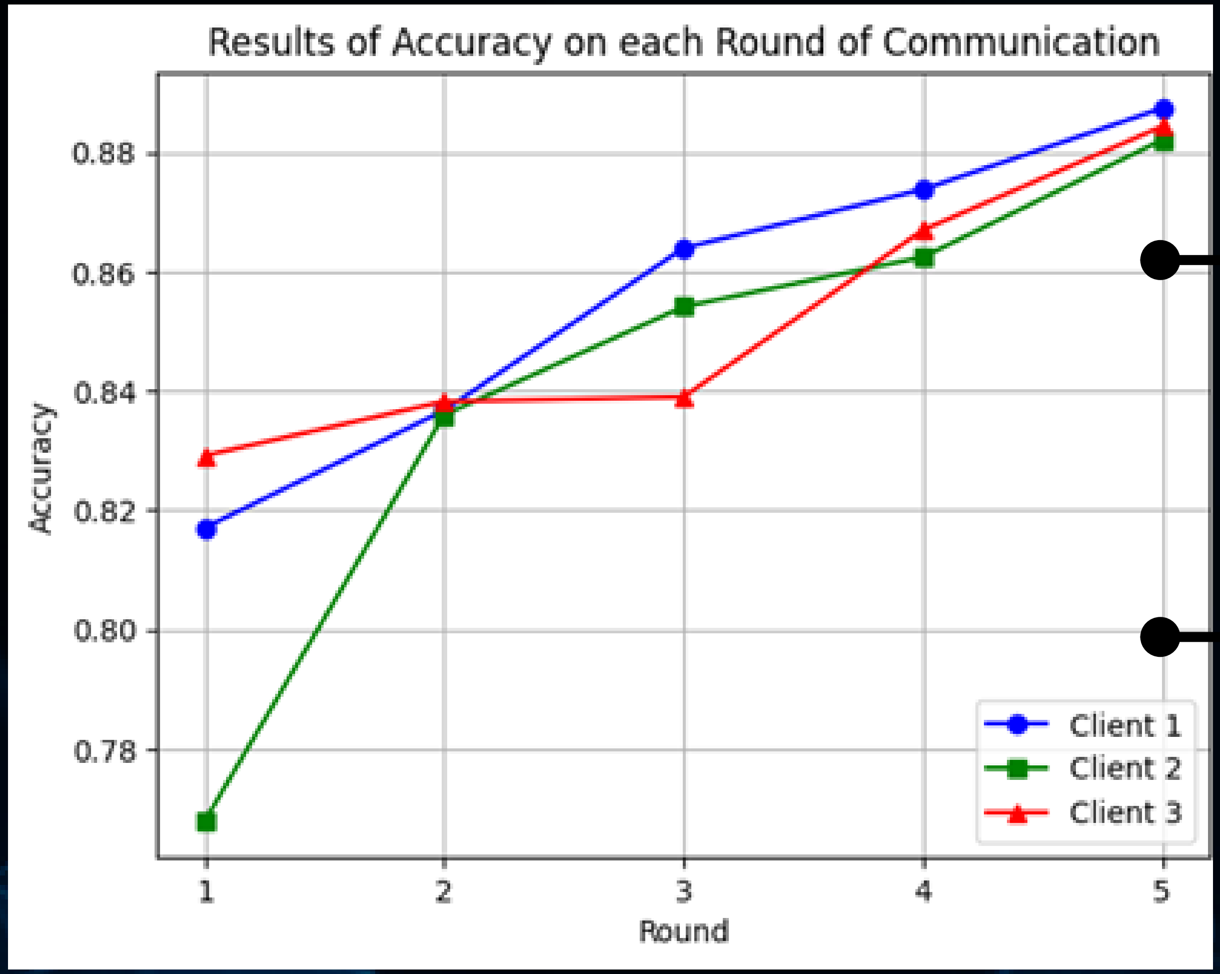
RESULTS OF THE FRAMEWORK USING PERFORMANCE BASED AGGREGATION

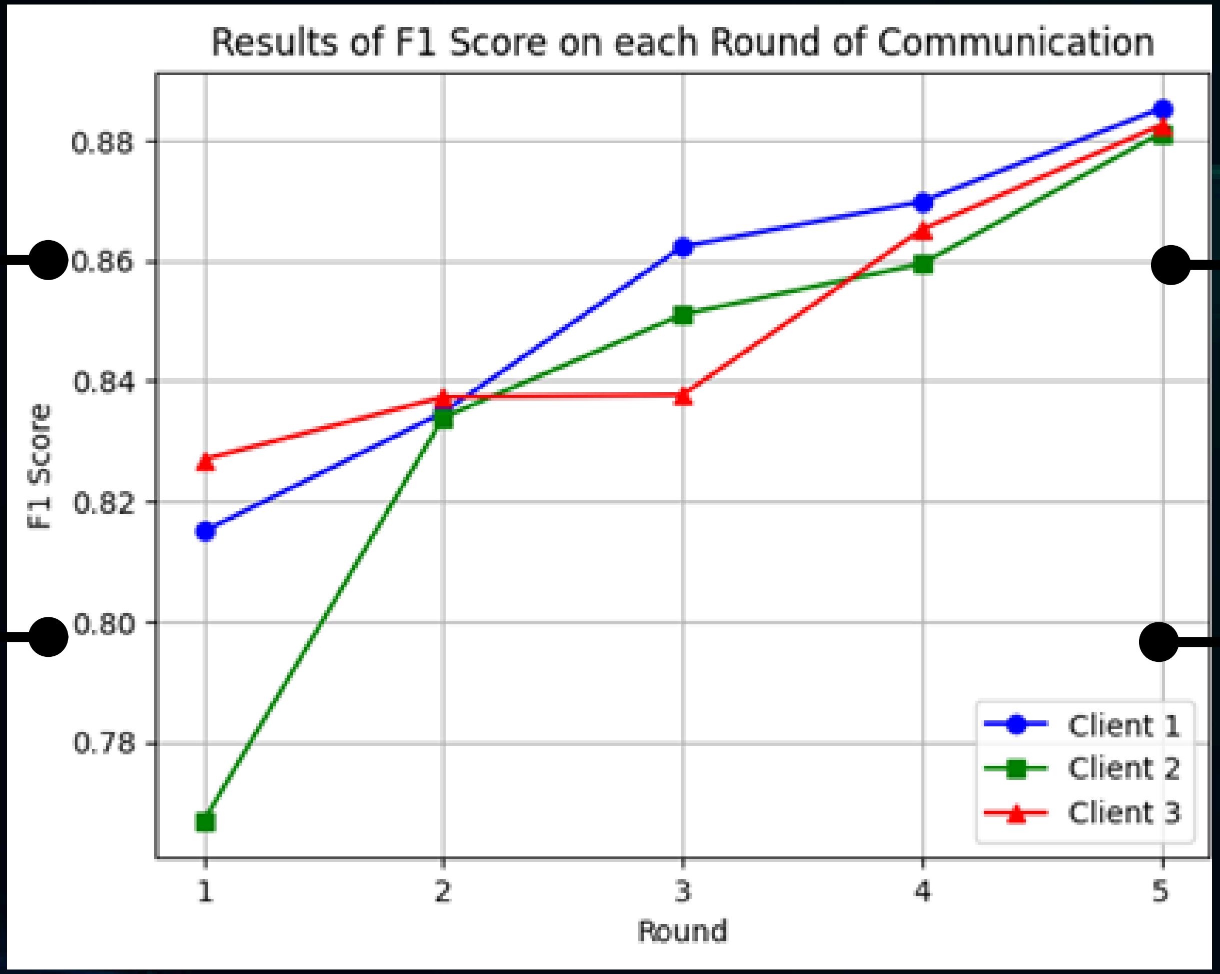


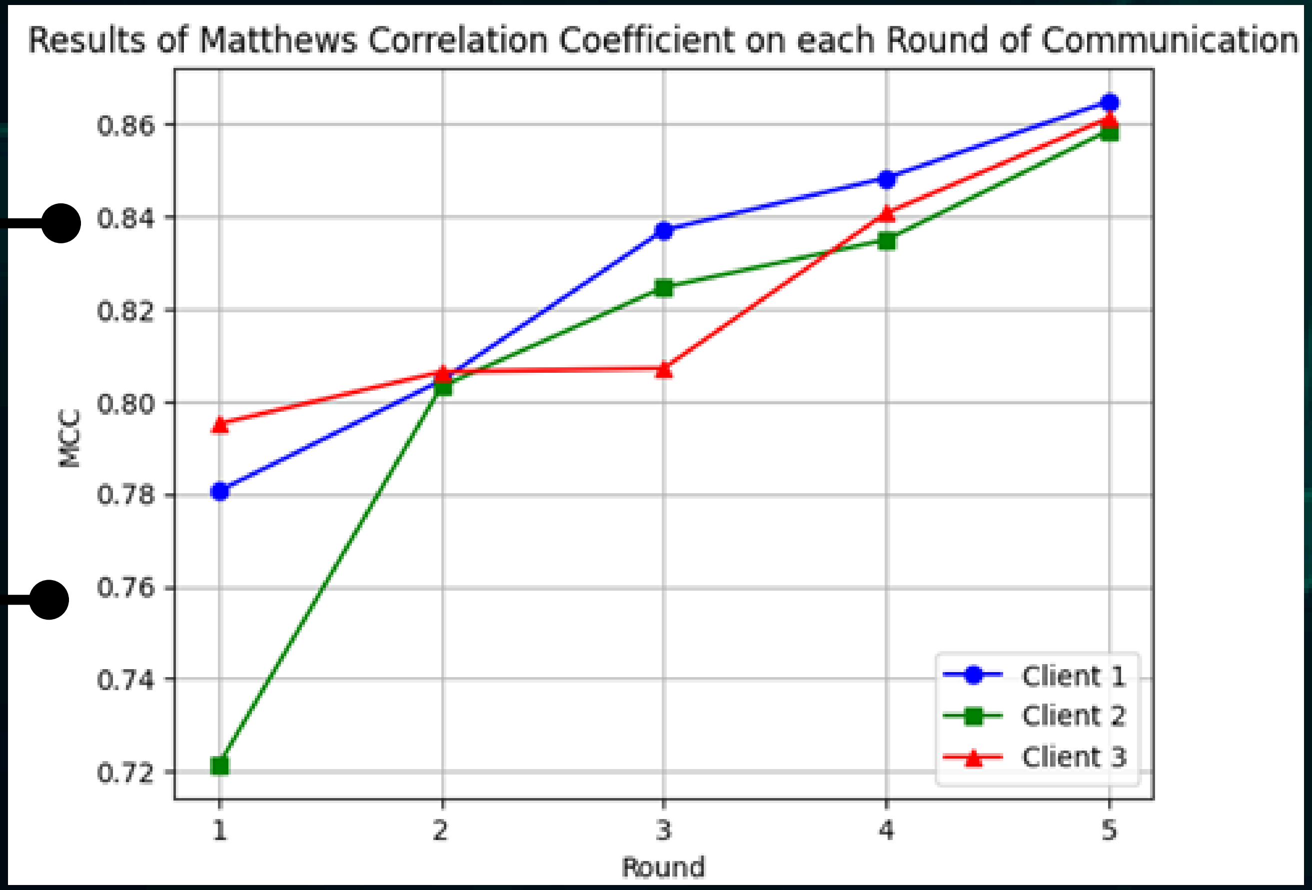




RESULTS OF THE FRAMEWORK USING DQ-FED AGGREGATION

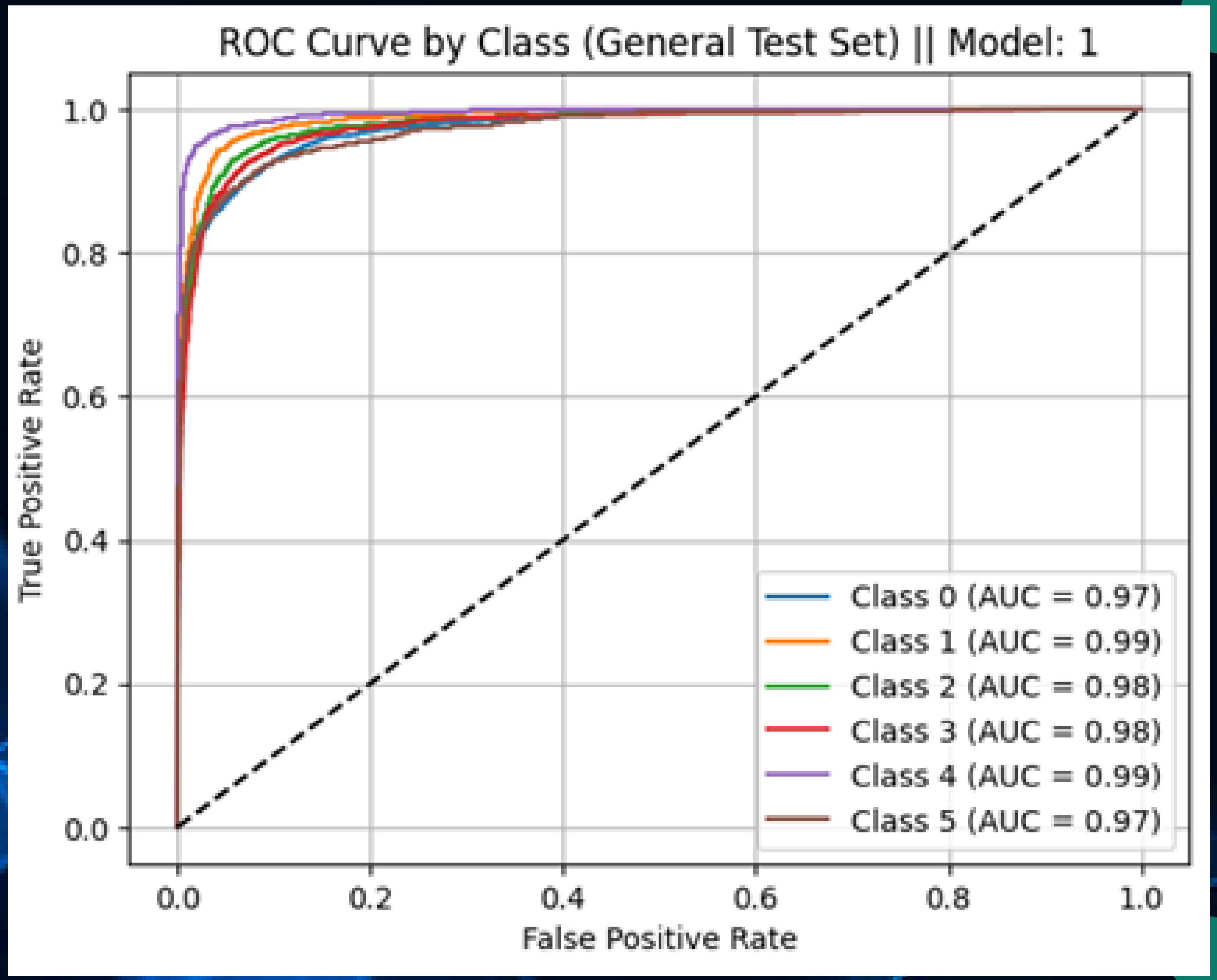


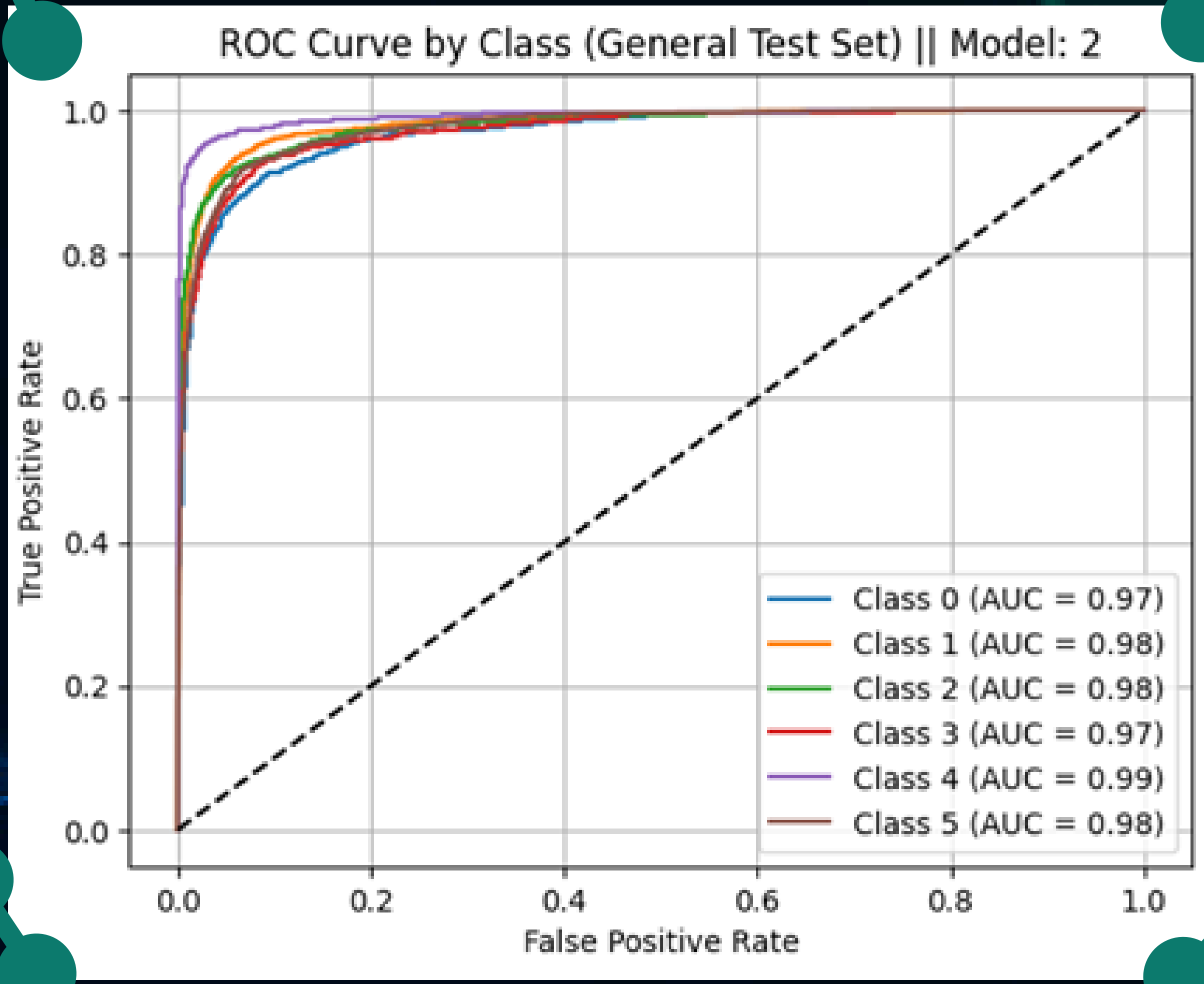


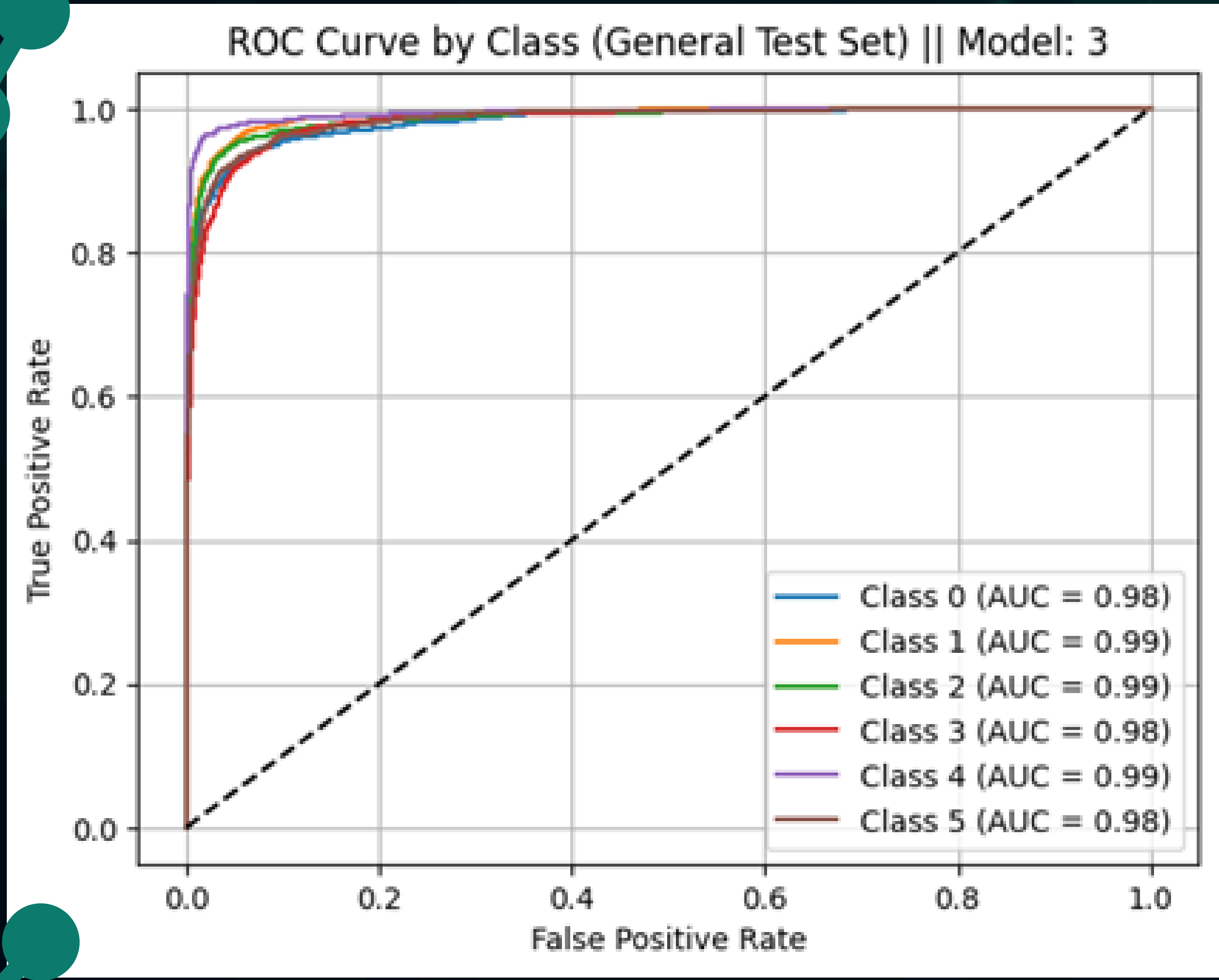


Client	Accuracy	F1-Score	MCC	Specificity
1	0,867	0,866	0,841	0,973
2	0,857	0,855	0,828	0,971
3	0,892	0,89	0,871	0,978
Average	0,872	0,870	0,847	0,974

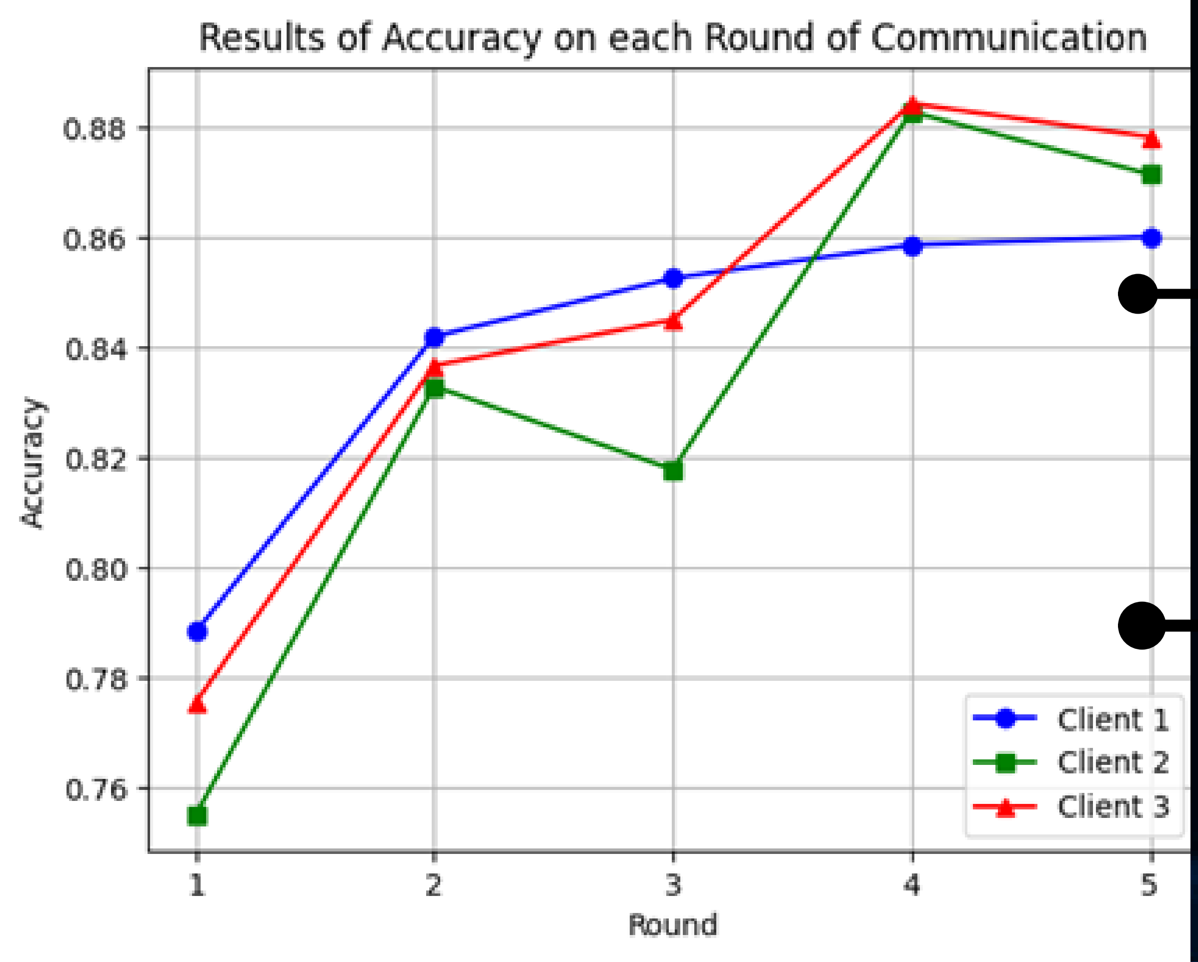
RESULTS OF THE FRAMEWORK USING DQ-FED AGGREGATION

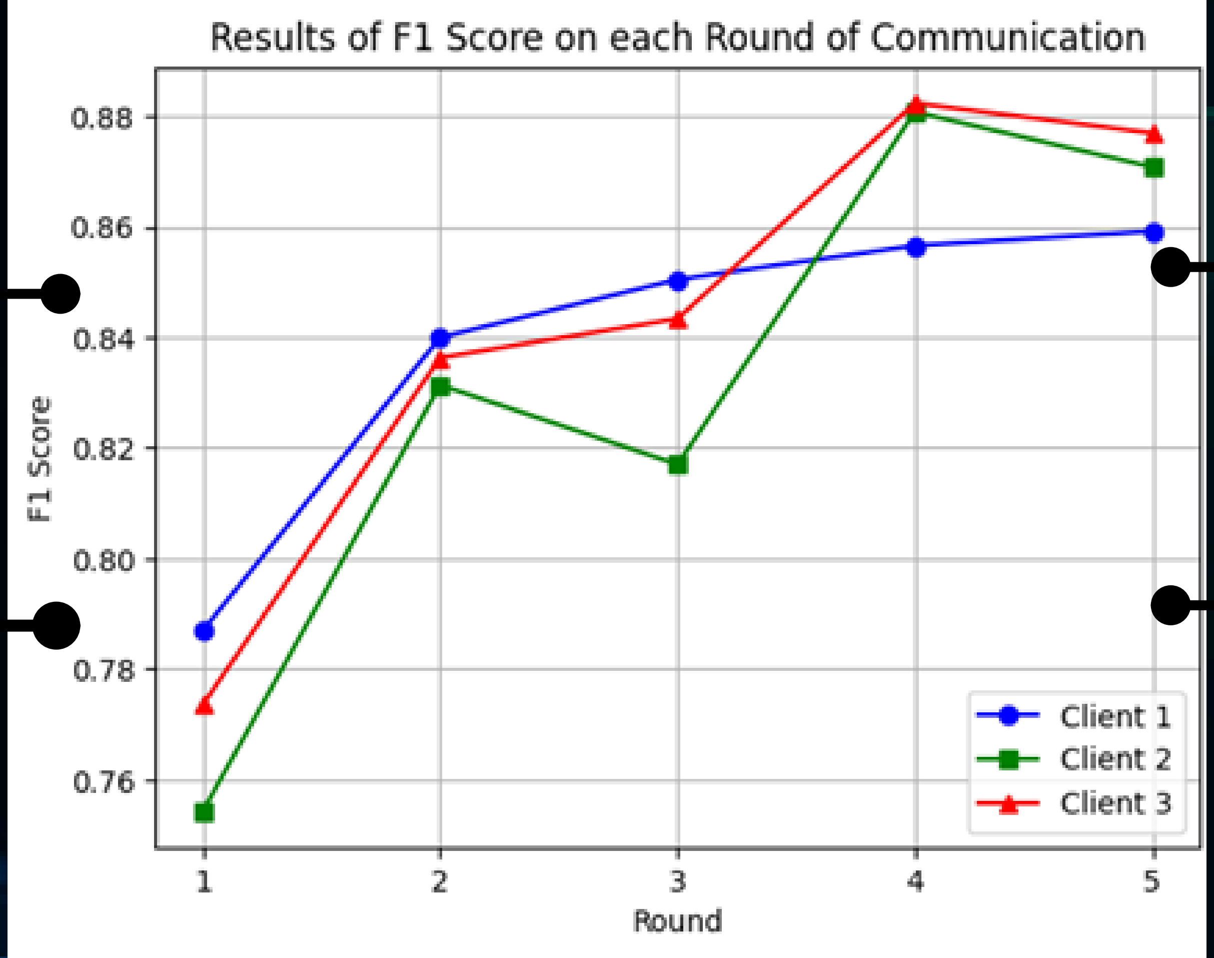


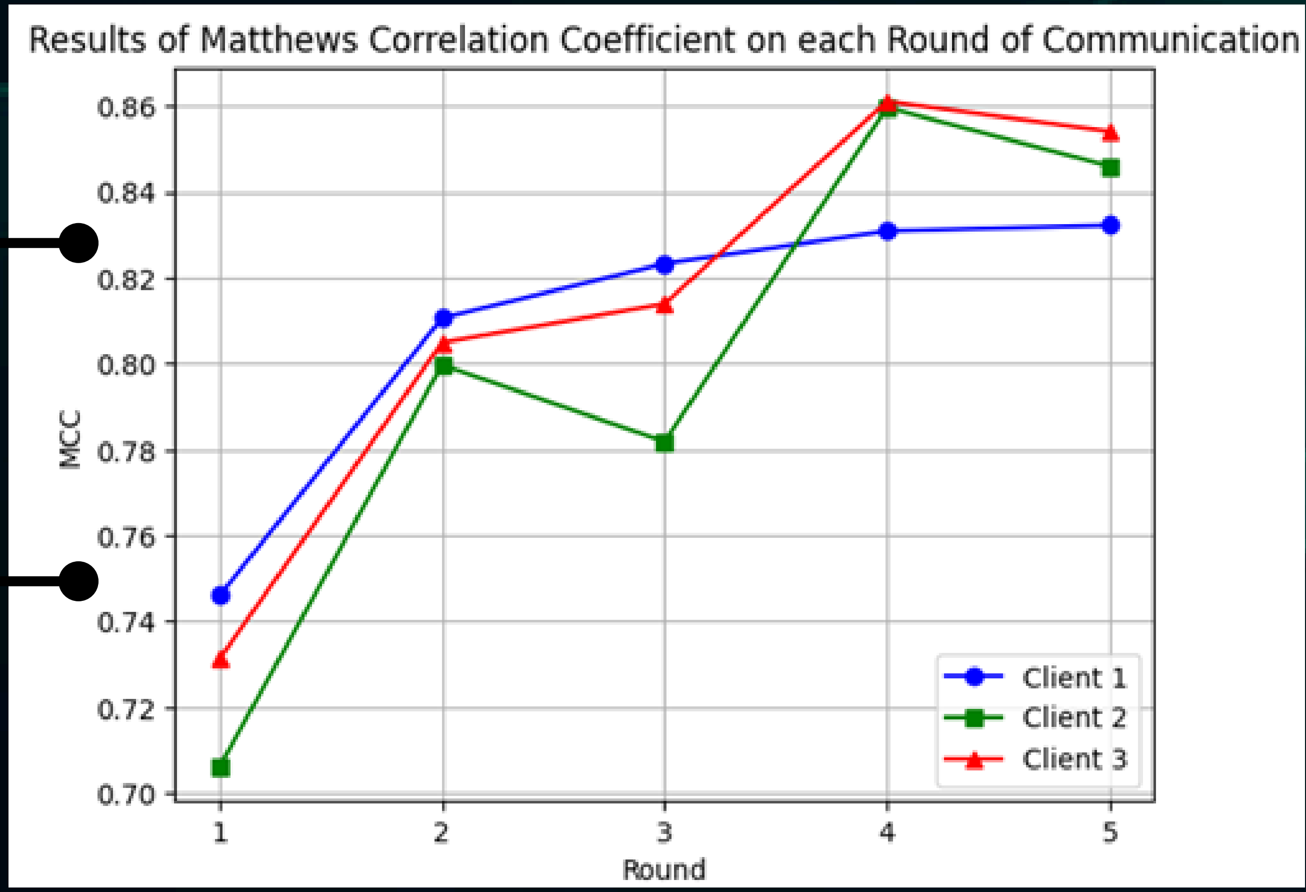




RESULTS OF THE FRAMEWORK USING FED-AVG AGGREGATION

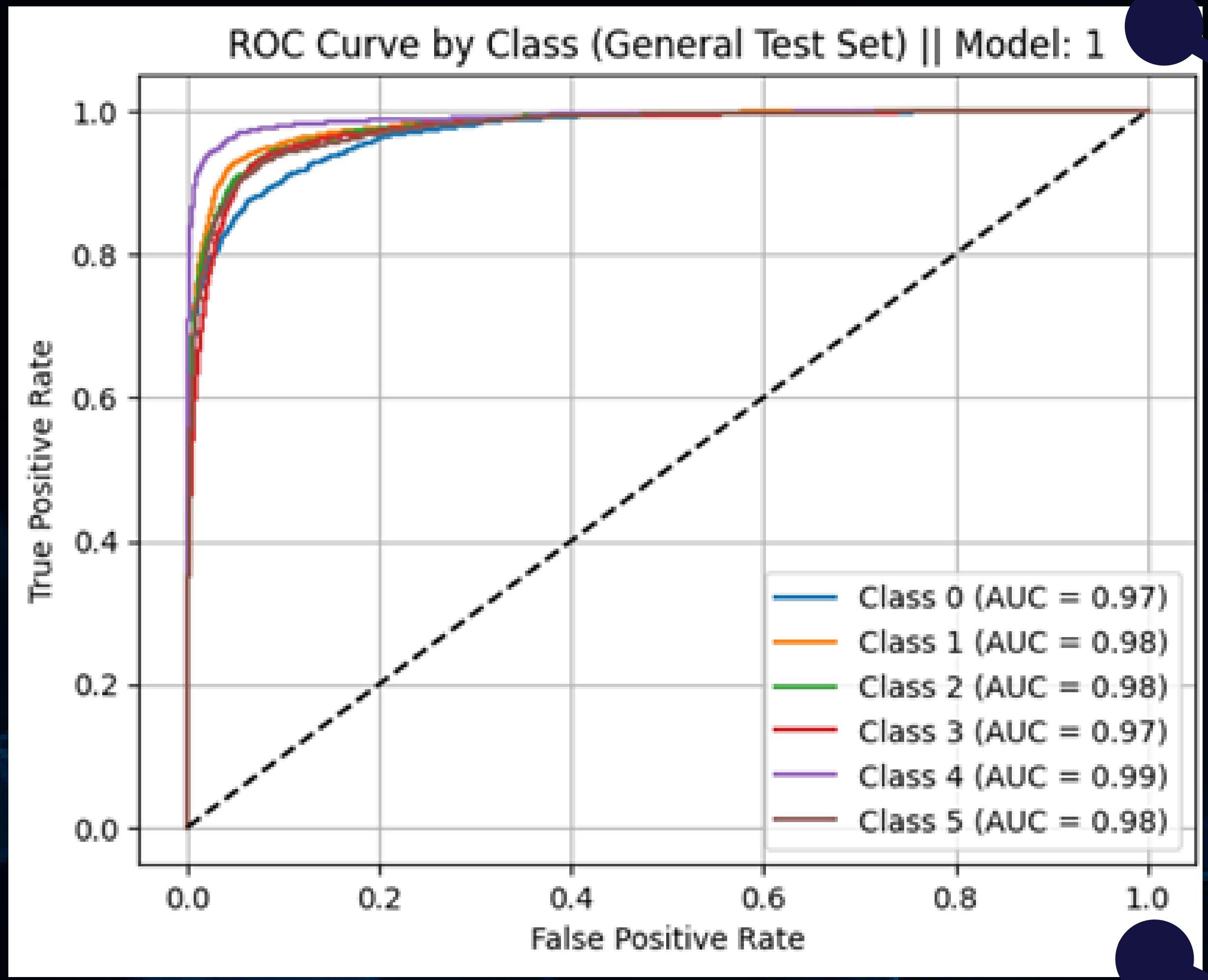


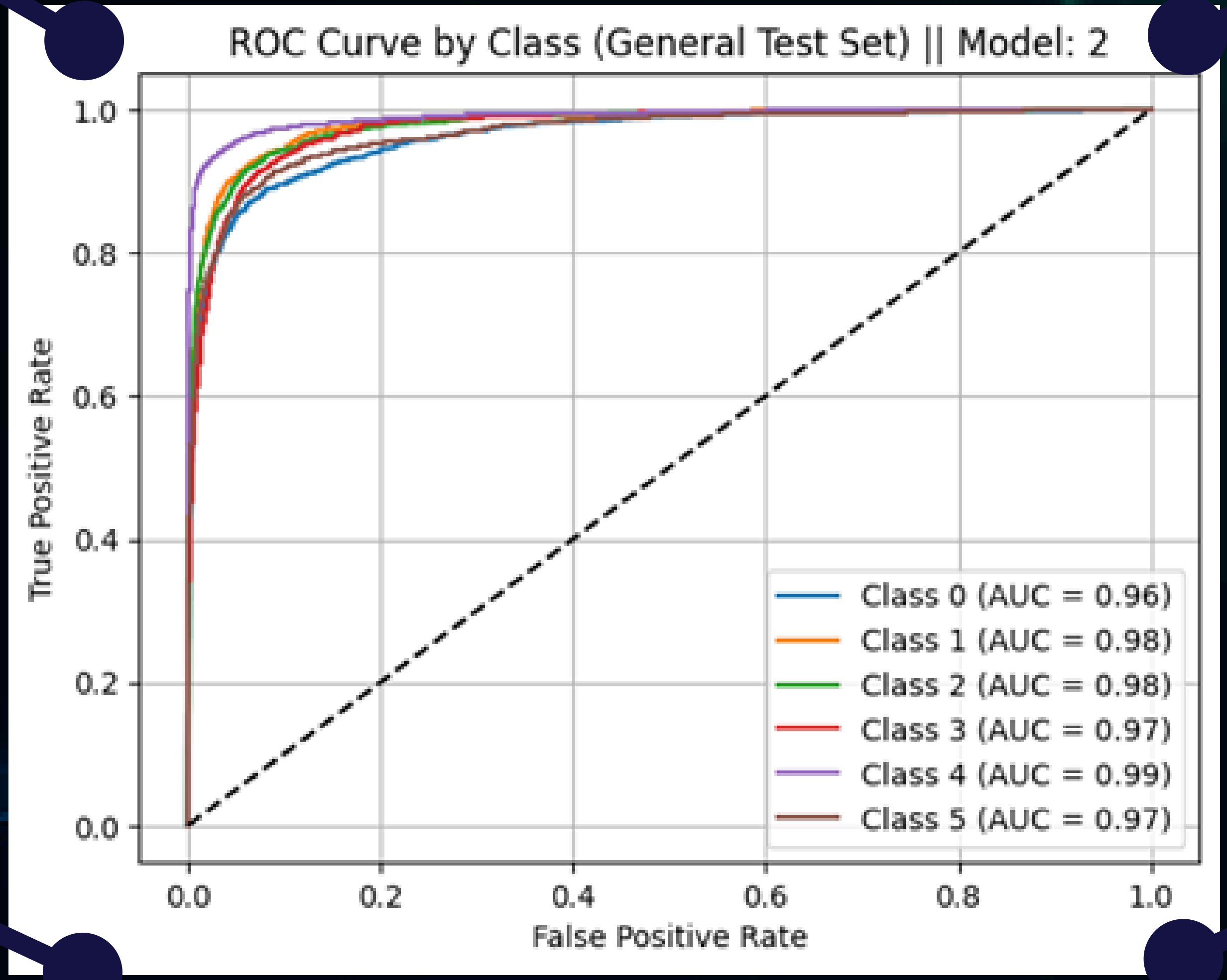


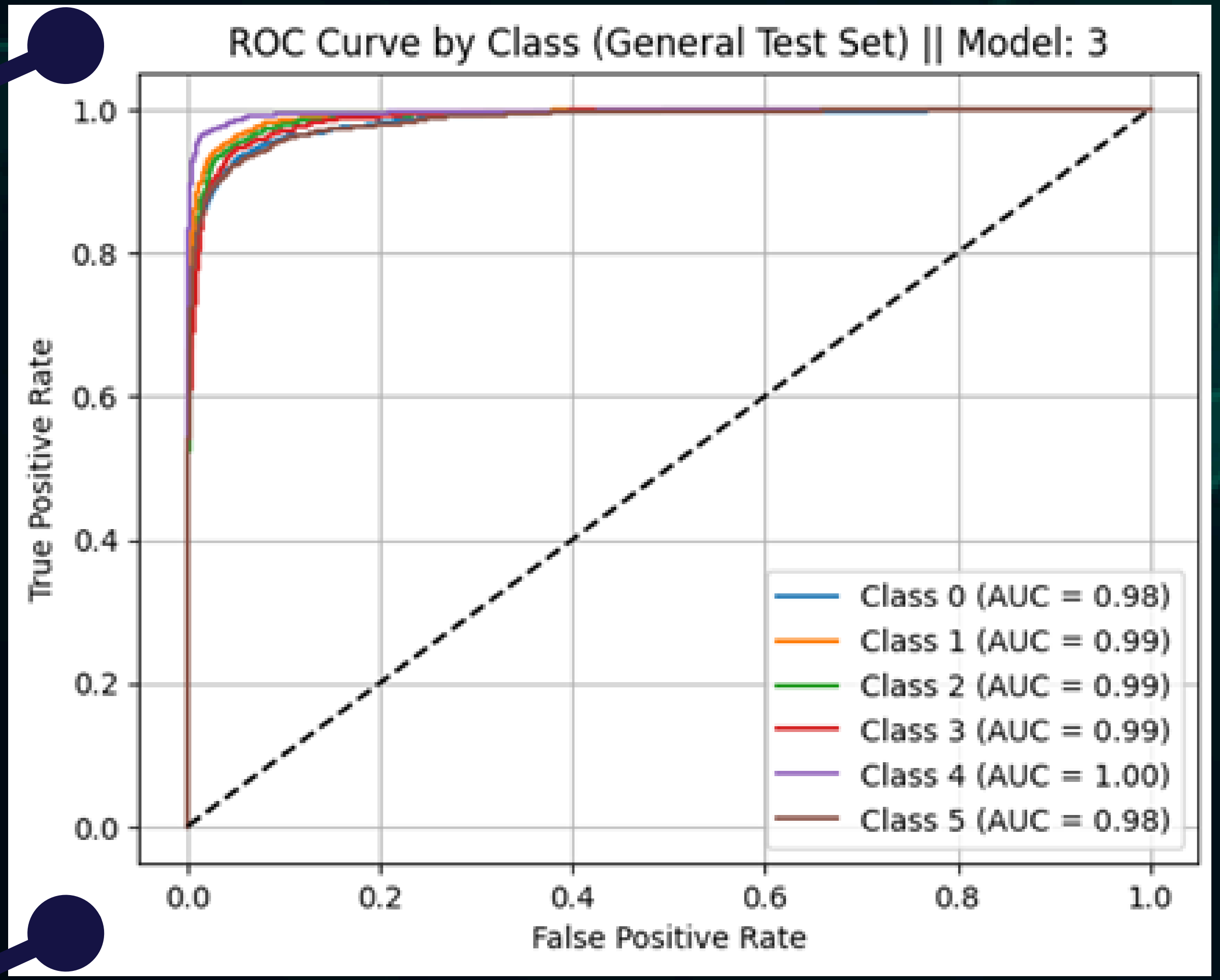


Client	Accuracy	F1-Score	MCC	Specificity
1	0,851	0,851	0,822	0,97
2	0,843	0,842	0,811	0,968
3	0,897	0,895	0,877	0,979
AVERAGE	0,864	0,863	0,837	0,972

RESULTS OF THE FRAMEWORK USING FED-AVG AGGREGATION





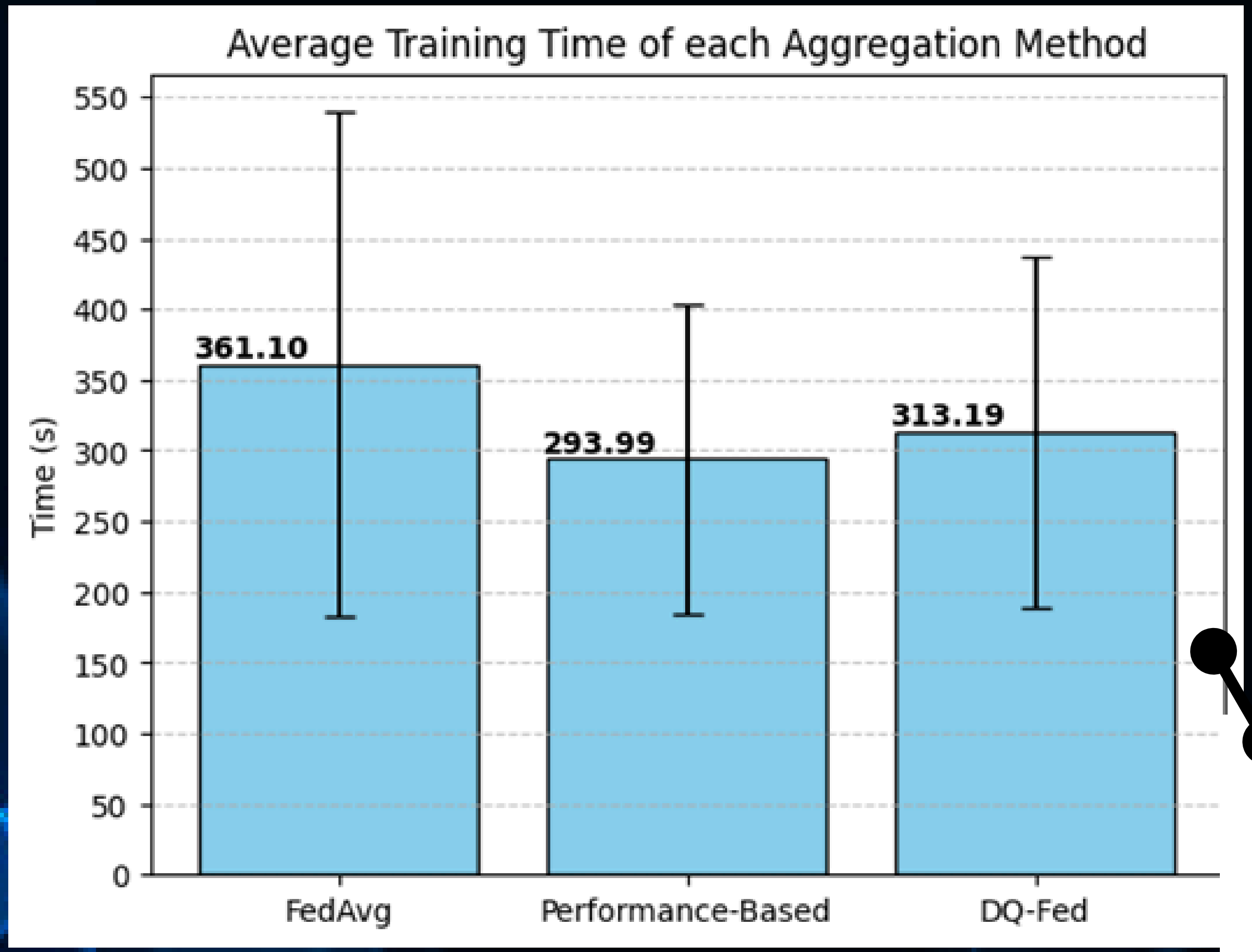


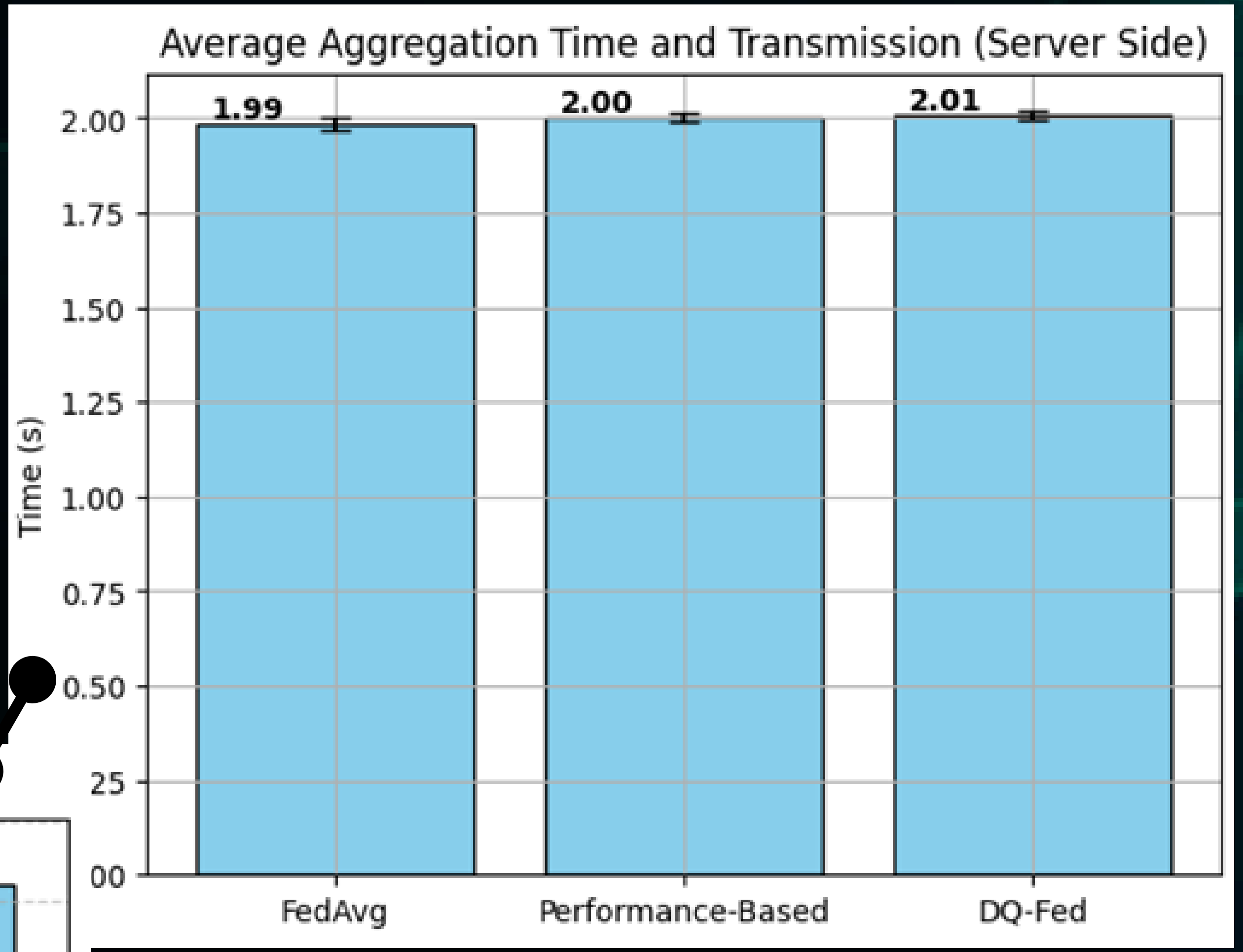
$$\frac{a_1 + a_2 + \cdots + a_n}{n}$$

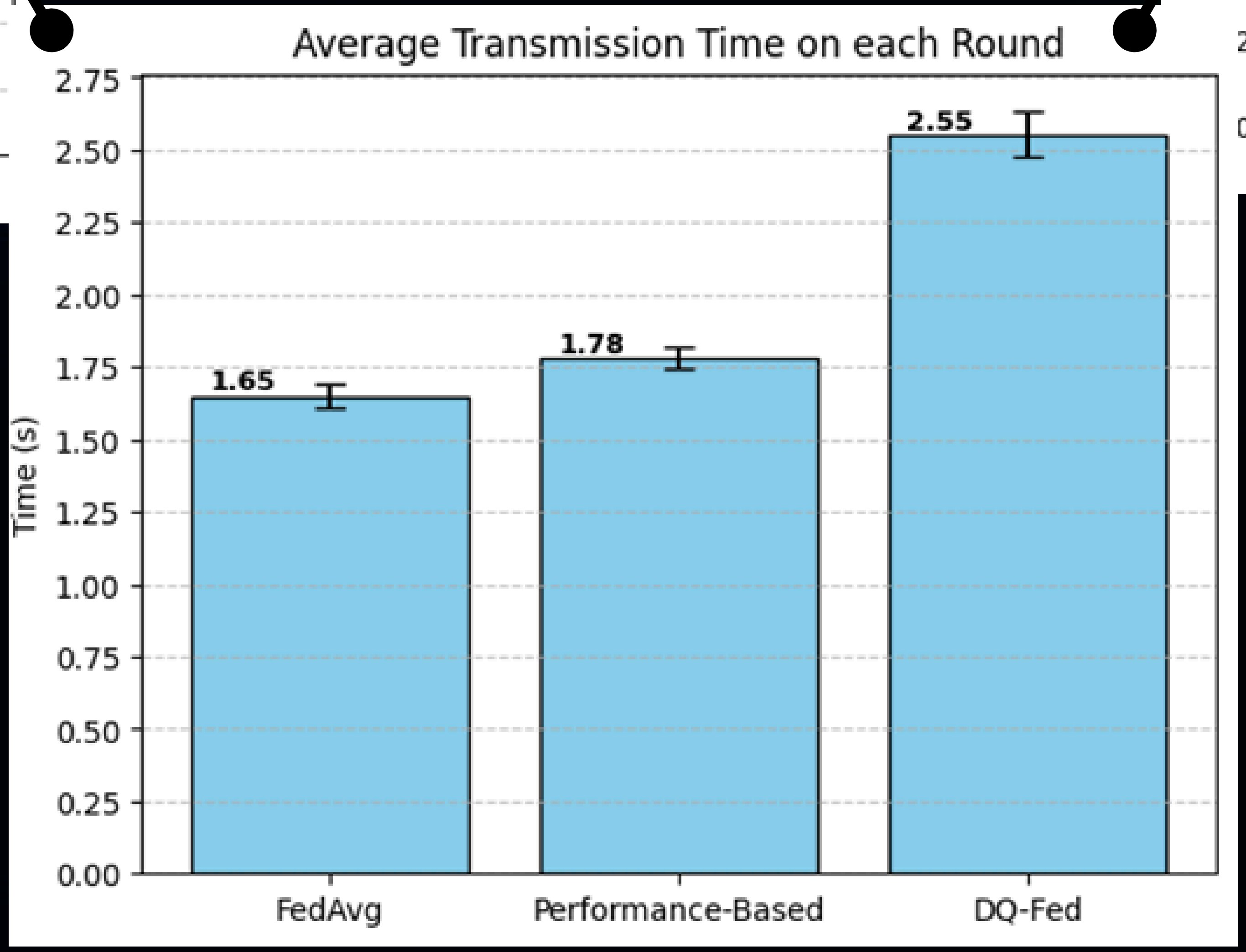
CONSOLIDATION OF CLASSIFICATION RESULTS

Federated Learning Method	Accuracy	F1-Score	MCC	Specificity	ROC-AUC
Performance-based	0,867	0,866	0,841	0,971	0,98
DQ-Fed	0,872	0,87	0,847	0,974	0,983
FedAvg	0,864	0,863	0,837	0,972	0,979

EXECUTION TIMES AND UPDATE LATENCY ASSESSMENT







CONCLUSIONS

- Generated a noise-filtered, balanced and representative cardiac dataset, scalable and depoyable for other Federated Learning instances associated to clinical uses.
- All aggregation methods analysed and compared achieved robust results (Acc $\geq$ 0.837, MCC = 0.811, Sensitivity = 0.968) with a minimal metric variation ranging 0.2 to 1.19%.
- DQ-Fed outperformed FedAvg (0.71%) and Performance-based (0.41%) in providing not only better results for classifying the six arrhythmias but also in promoting stability, adaptability and reduced overfitting.

CONCLUSIONS

- FedAvg has been the alternative with the lowest classification scores and slowest optimization, however, its only advantage is that ensures a faster communication between client-server.
- The DQ-Fed method showed the slowest transmission time (2.55 s), however, this accounted for less than 0.02% of the total round duration, thus maintained a real-time feasibility with minimal delays within the client local process as well as the interaction with the server.
- As a proposal for a future work and continuation of the present project, would be including protocols that ensure the framework to be robust against situations that can be found in clinical and operational deployments such as unstable connectivity, limited computational resources or access to basic services.

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APPENDIX

Link to Video Demo:

https://drive.google.com/file/d/1Zp7xRugkFdG6VpLdHQ-B9Ts_bPalFO1J/view?usp=sharing

THANK YOU!

FOR YOUR ATTENTION